

PROIECTAREA CU MICROPROCESOARE

C3 2025-2026 Sem. 2

Proiect 7

Sapt 7

8 aprilie 2026

19:00-20:00

serban@upit.ro

Regulament disciplina

Nota finala este formata din activitatile:

- Laborator 15%
- Lucrare control 15%
- Prezentare si sustinere proiect 20%
- Examen 50%
- Bonus prezenta – 10%

Conditii pentru promovare:

- Nota de la laborator trebuie sa fie minim 5 (prezenta obligatorie la toate sedintele de laborator);
- Nota proiect minim 5 (conditie obligatorie pentru intrare in examen);
- Nota de la examen trebuie sa fie minim 5;
- Nota de minim 5 la Lucrarea de control.

In cazul reluarii disciplinei intr-un alt an universitar, activitatile nepromovate trebuie parcurse din nou.

Cerinte proiect

1). Realizarea suportului scris (format pdf) care sa contina capitolele:

Proiectarea Hardware si Proiectarea Software

- In cap. Proiectarea Hardware trebuie sa fie inclusa argumentarea pentru logica de conectare a circuitelor de memorie si I/O, respectiv a perifericelor;
- Este **obligatorie** schema de conectare finala a tuturor componentelor (eventual fara circuitele de alimentare) cu precizarea tuturor pinilor circuitelor folosite si a modului de conectare a acestora; va fi prezentat modul de programare a tablei prin folosirea tastelor (manualul de utilizare);
- In cap. Proiectarea Software trebuie sa fie inclusa logica de constructie a programului software cu toate componentele sale (prin descrieri explicative si organigrame), respectiv listingul programului;
- Este **obligatorie** lista de variabile folosite sub forma unui tabel de forma:

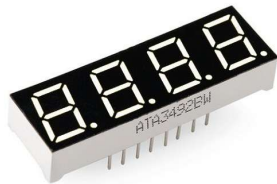
Nume variabila	Semnificatie	Valoarea de initializare	Mod de reprezentare	Adresa de stocare in memorie RAM

- Folosirea unui simulator pentru programul (sau portiuni din el) realizat este un bonus.

2). Predarea proiectului se va face pana in ziua de **23 mai 2026 ora 23:00** pe platforma moddle.

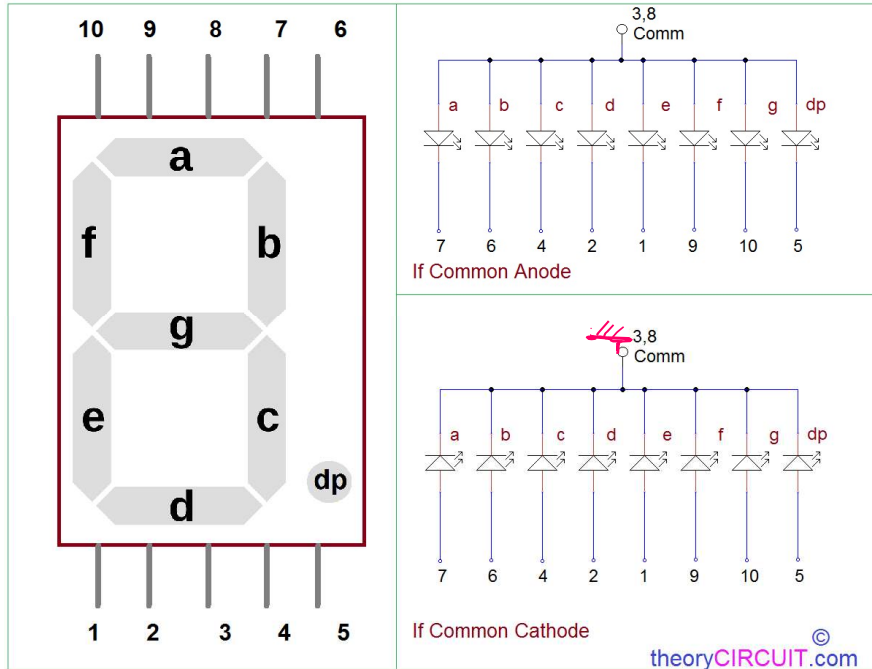
3). Sustinerea proiectului de catre realizator (activitate obligatorie) printr-o sesiune de intrebari legate de solutiile folosite (in ultima saptamana din semestru 25 – 29 mai 2026 la o data stablita de comun acord).

Celule de afisaj LED cu 7 segmente



Celule de afisaj LED cu 7 segmente

7 Segment Display Pinout

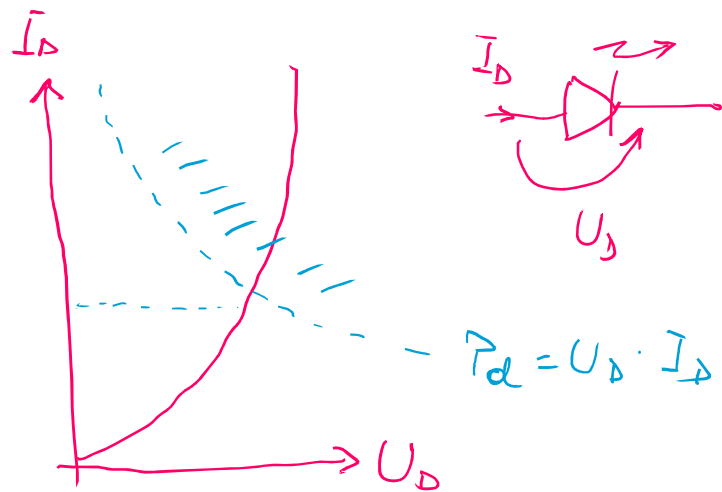
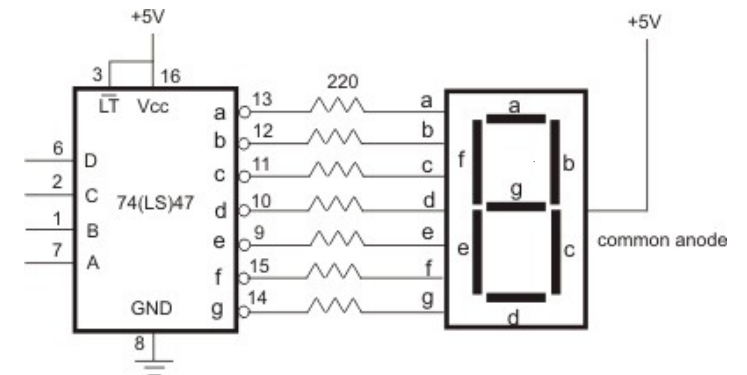
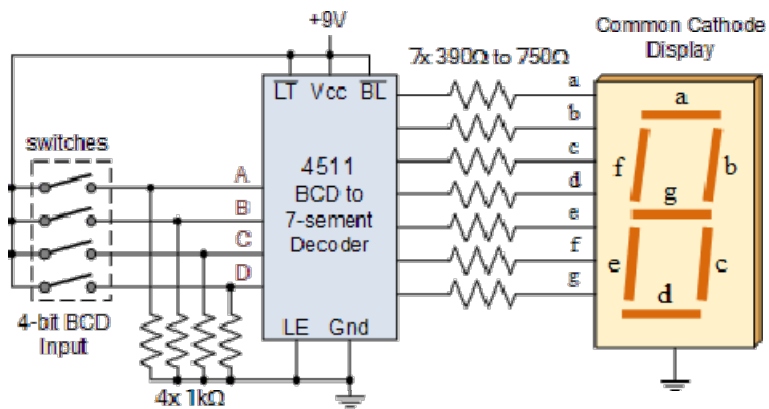


CATHODE COMMON

segment outputs							display
a	b	c	d	e	f	g	
1	1	1	1	1	1	0	0
0	1	1	0	0	0	0	1
1	1	0	1	1	0	1	2
1	1	1	1	0	0	1	3
0	1	1	0	0	1	1	4
1	0	1	1	0	1	1	5
0	0	1	1	1	1	1	6
1	1	1	0	0	0	0	7
1	1	1	1	1	1	1	8
1	1	1	0	0	1	1	9

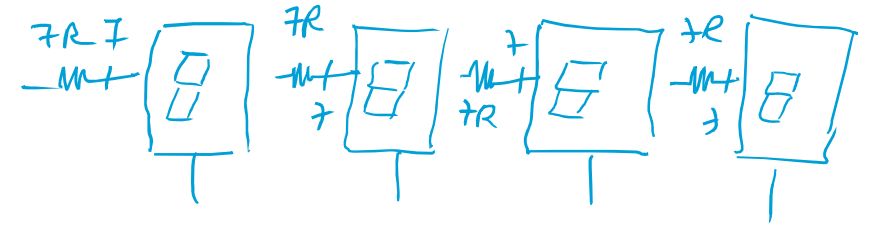
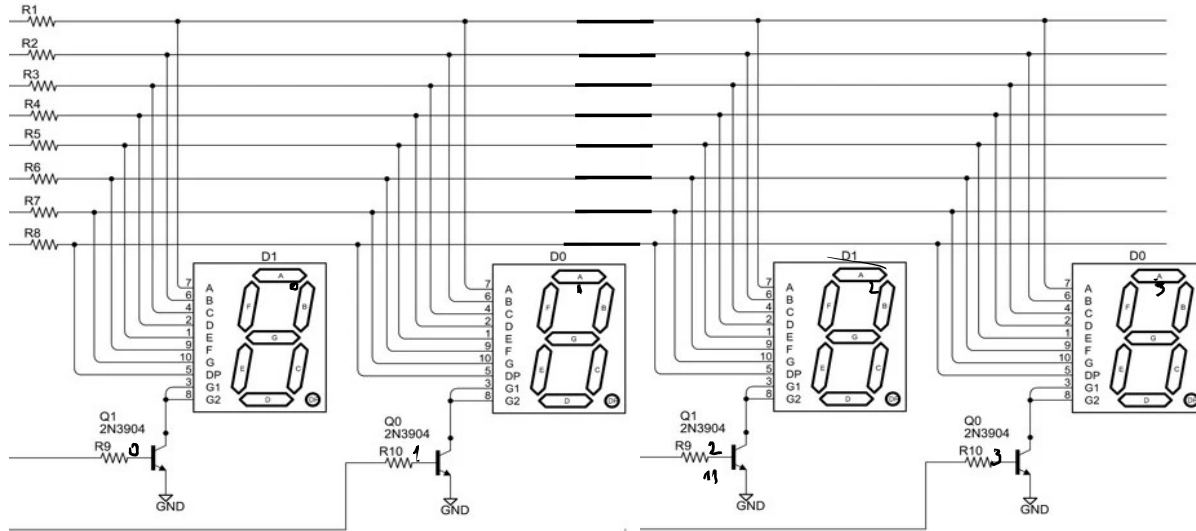
7-Segment Display

Celule de afisaj LED cu 7 segmente



comanda clasic pentru
celula LED 7 segmente.

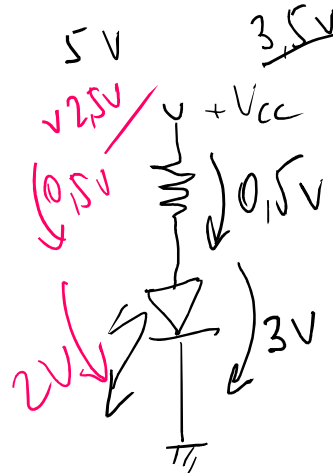
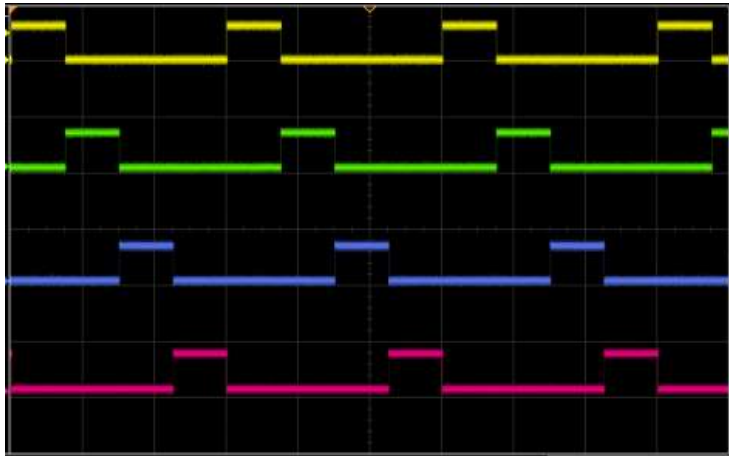
Proiectarea afisajului cu celule LED 7 segmente comandat prin multiplexare



$7R \times 4 = 28R$ Rezistente
 $7 \times 4 = 28$ fire comanda
 4 - catzi

(comanda clasic)

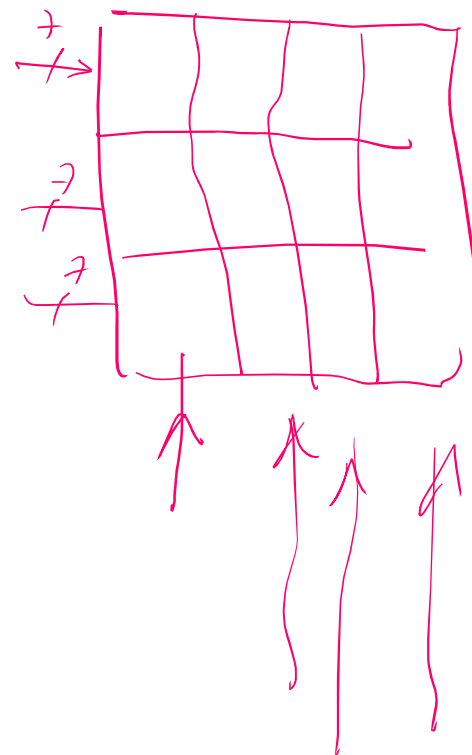
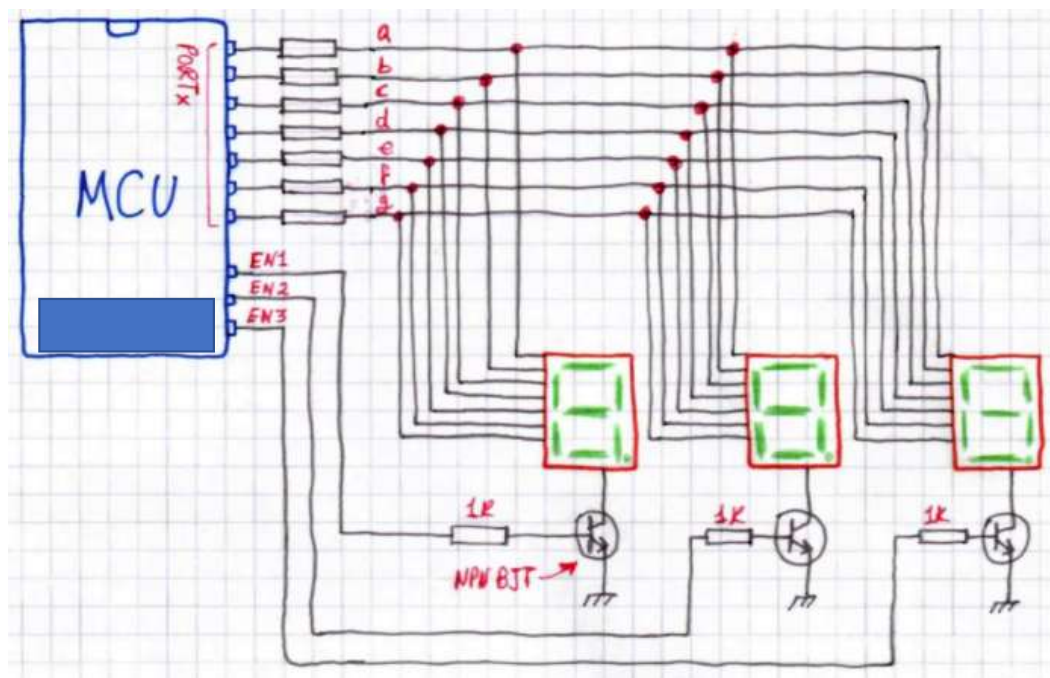
$t = ?$ $f > 50 \text{ kHz}$ $T < 20 \text{ ms}$

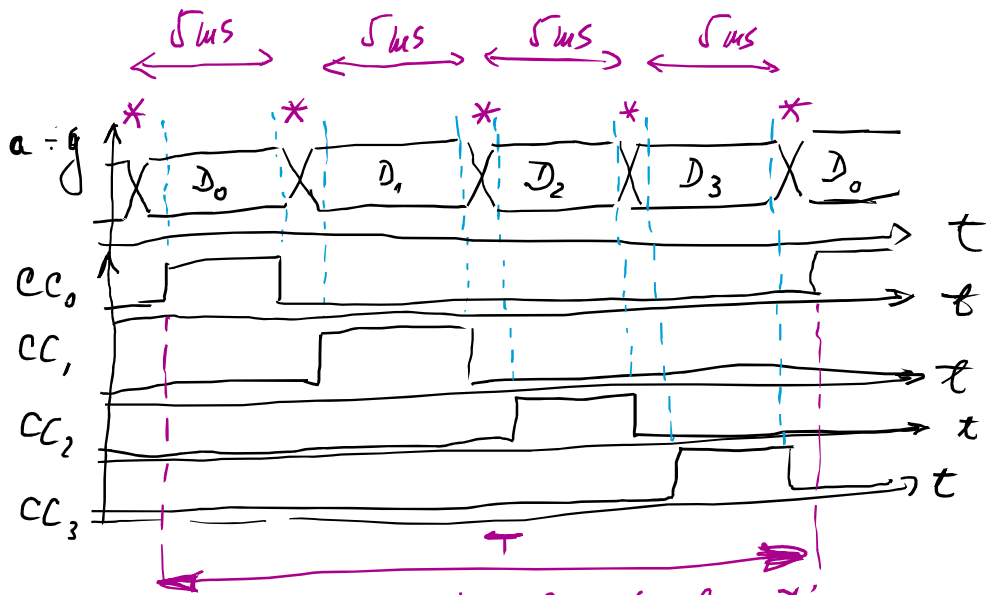


Comanda prin multiplexare
 Resurse

- $7R$ - rezistente
 - $7 + 4 = 11$ fire comanda

Proiectarea afisajului cu celule LED 7 segmente comandat prin multiplexare



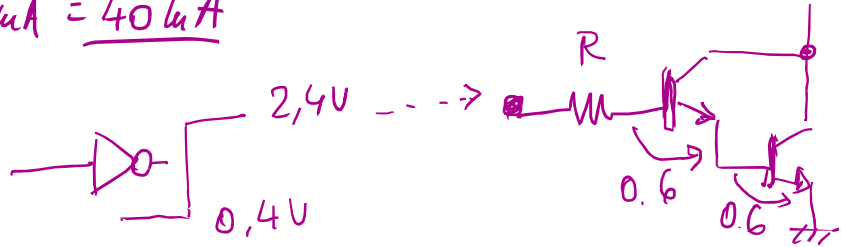


$T \leq 20 \text{ ms}$ ($f \geq 50 \text{ Hz}$) $n = 4$ digits
 $T/n = \frac{20 \text{ ms}}{4} = 5 \text{ ms} \Rightarrow$ necesitate folosire timer

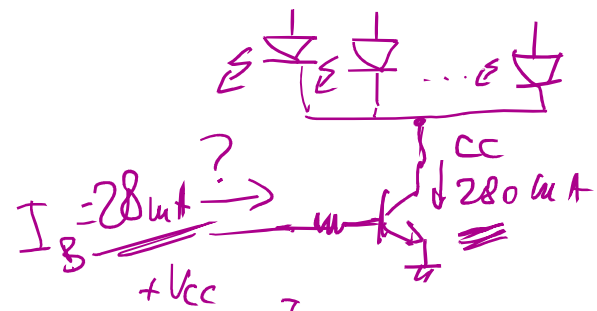
$\Rightarrow$ semnalizari la 5 ms prin cereri de intrerupere catre CPU

$I_{\text{peak}} \geq n \cdot I_D = 4 \cdot 10 \text{ mA} = 40 \text{ mA}$

$\Rightarrow$ necesitate circuito amplificare curent



Circuitul de amplificare in curent Ct_2 (cotelod comun) -



medie putere BD 141, 143

$$V_{CC} = I_D \cdot R + V_D + U_{CE}$$

$$5 = 40 R + 2 + \frac{U_{CE \text{ SAT}}}{1 \text{ V}} \quad (2N2222)$$

$$R = \frac{5 - 2 - 1}{40} = \frac{2}{40} = 0,05 \text{ k}$$

$$P_d = U_{CE} \cdot I_C$$

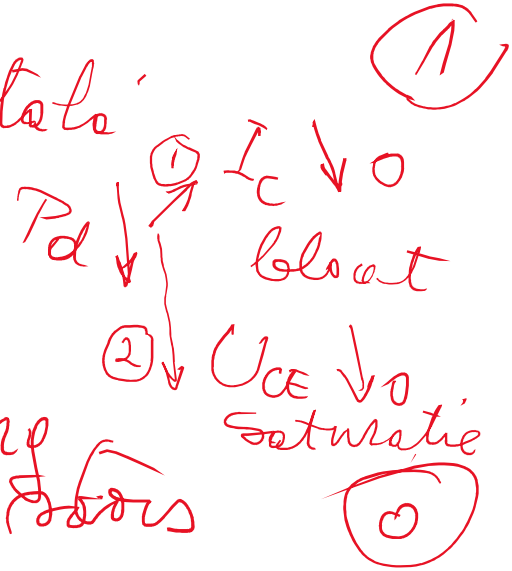
$$\beta_{\text{SAT}} = 101 \quad = 50 \text{ R}$$

Regim saturat $U_{CE} \rightarrow 0$
 Regim blocat $I_C \rightarrow 0$

Operare cu tranzistori in logica digitala

1) Putere disipata-

$$P_{d\text{tranz}} = U_{CE} \cdot I_C$$



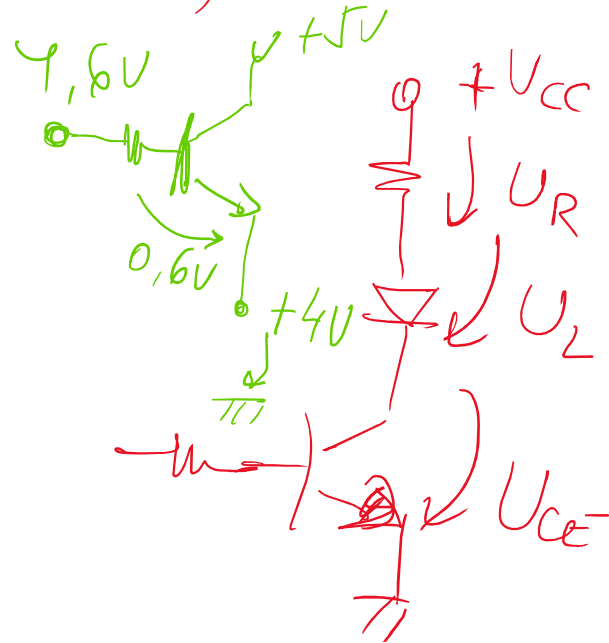
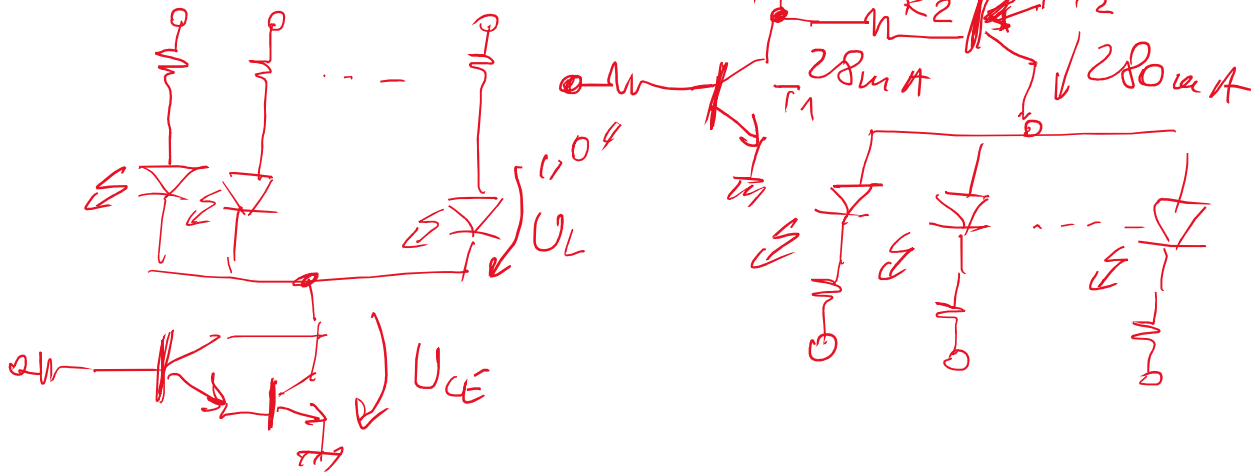
Switching transistors

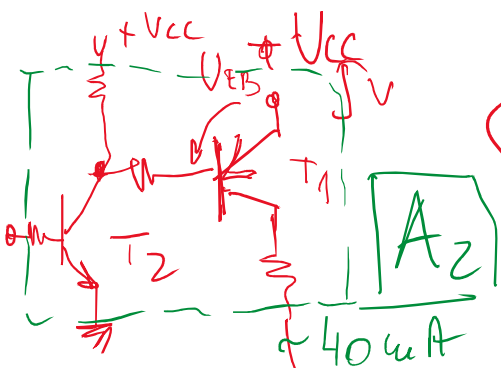
$$P = U \cdot I = I \cdot R \cdot I = I^2 \cdot R$$



Catod comun

anod comun

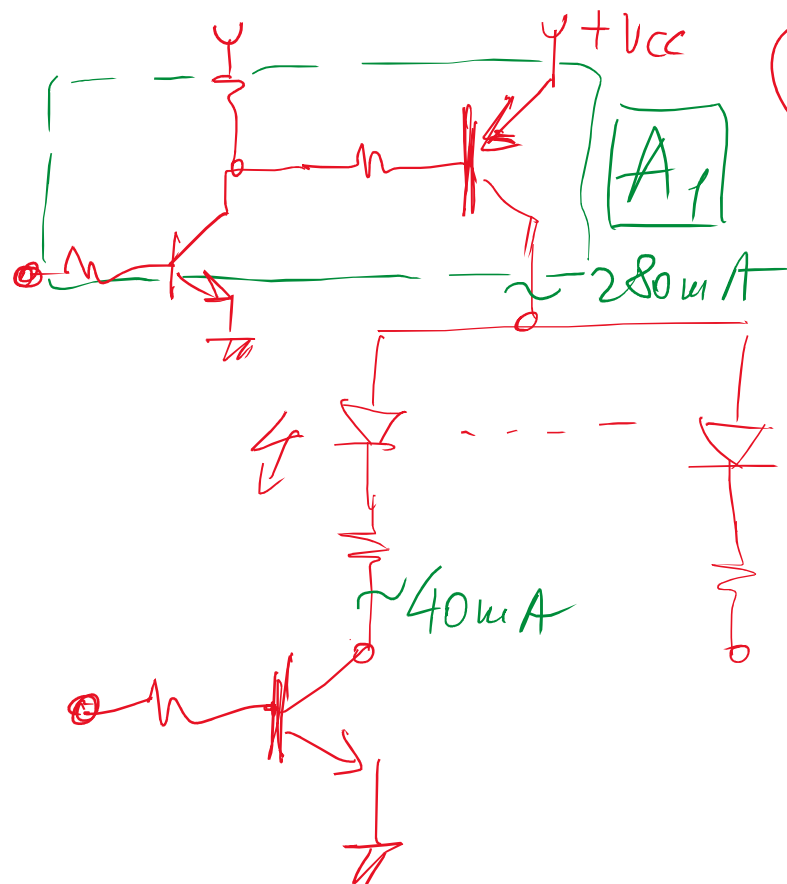
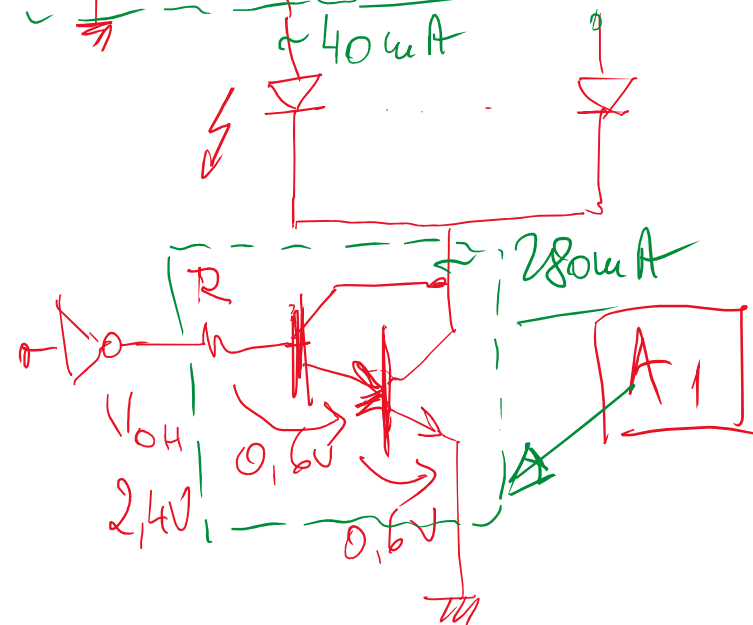




(CC)

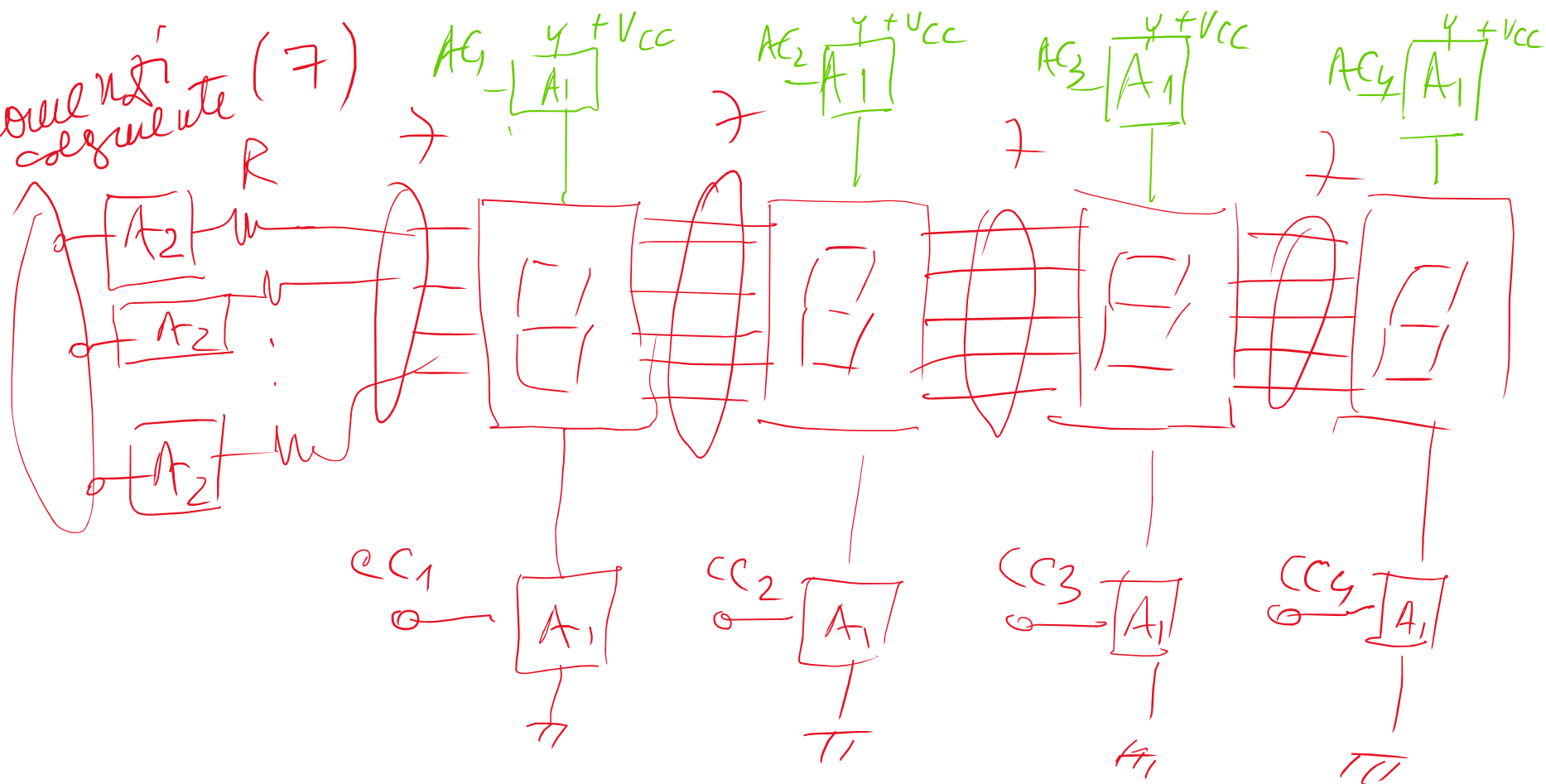
$$V_{OL} \leq 0,4V$$

$$V_{OH} \geq 2,4V$$



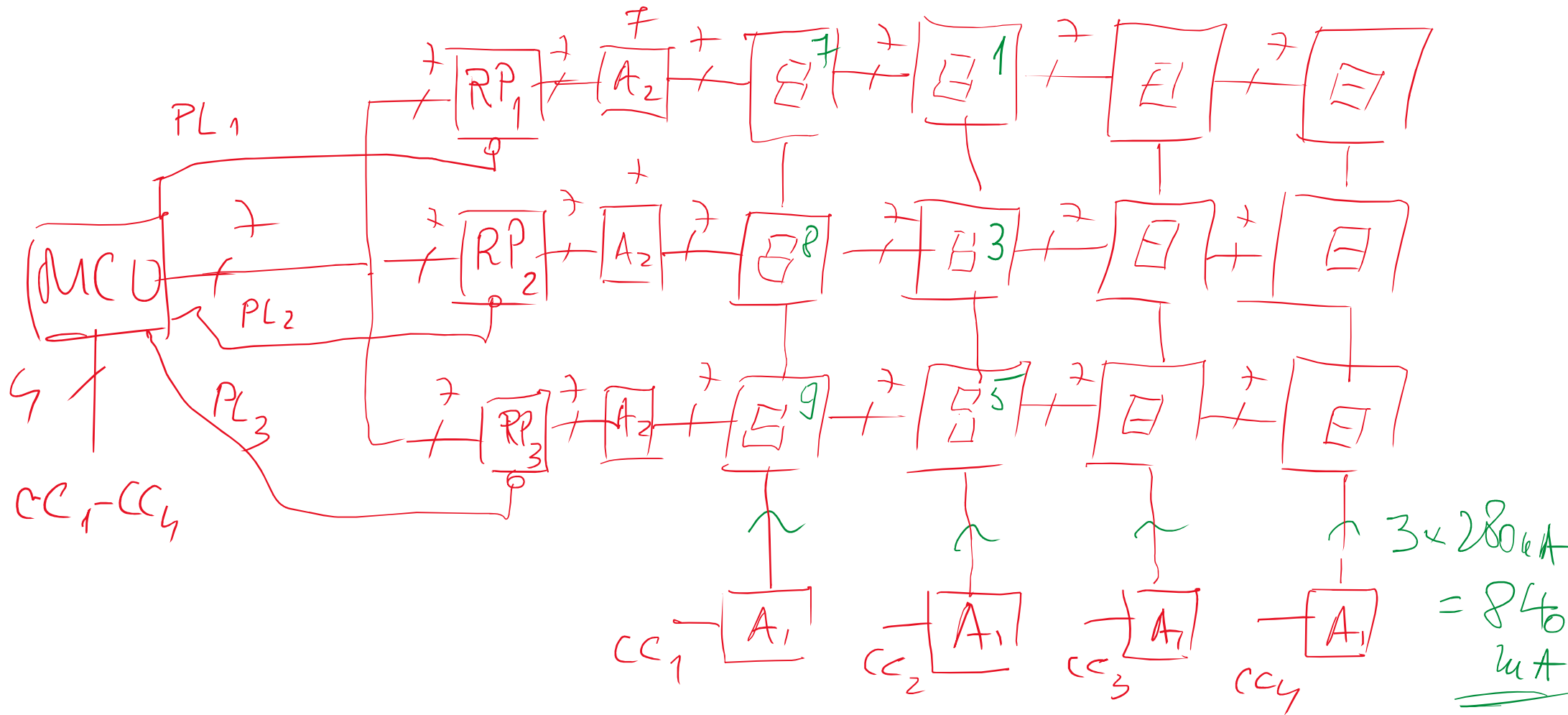
(AC)

could not
calculate (7)



AE

PL-parallel load

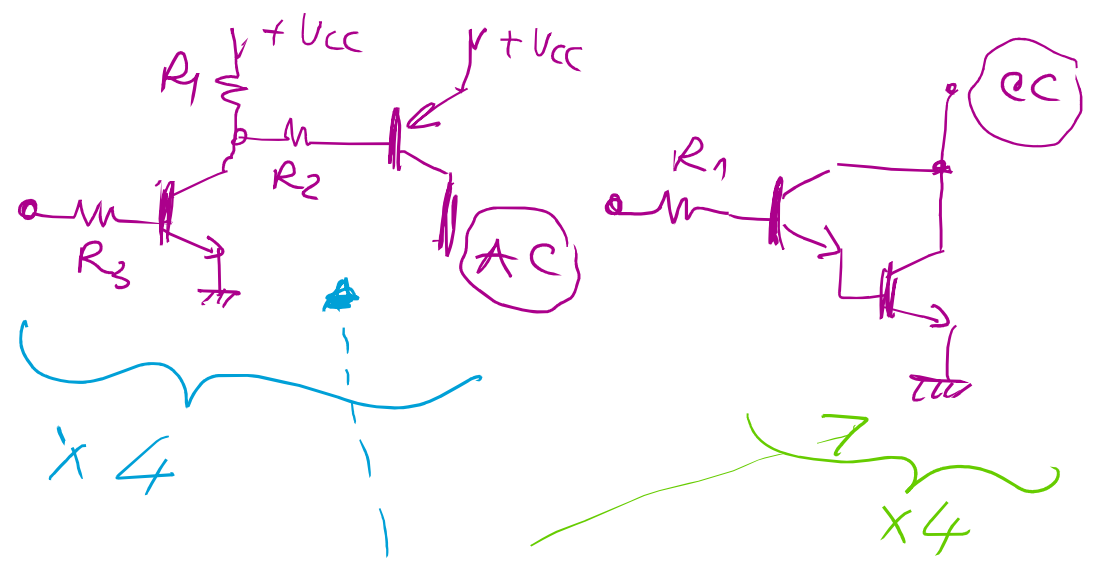
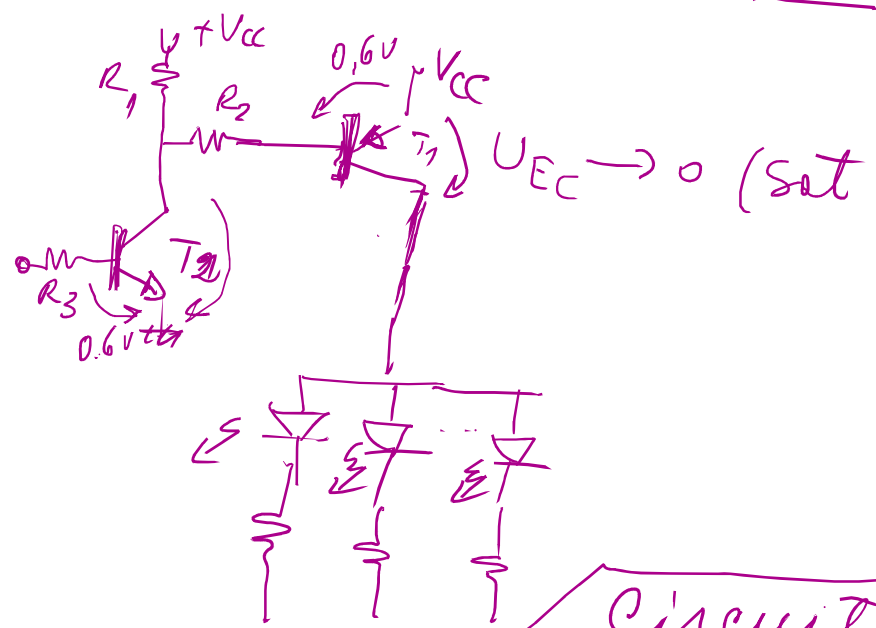


SURSE

- 1) Căutare celule LED 7seg (www.farnell.com)
- 3) Căutare tranzistoare (—||—) pentru A_1 și A_2
- 2) Scheme principale de multiplexare pentru matricea de celule 7seg din tablă (calcul curent)
- 4) Proiectarea A_1 și A_2 (valori de rezistențe + alegere tranzistor)

CA2

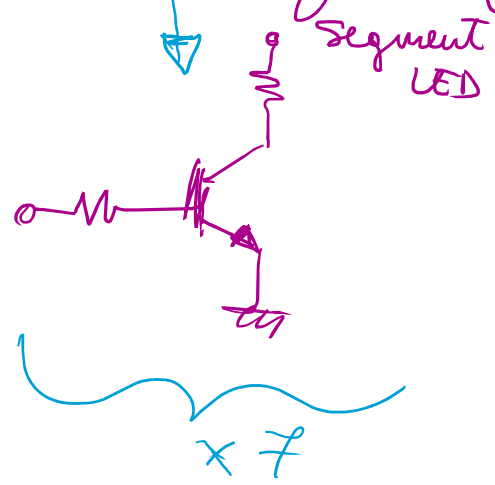
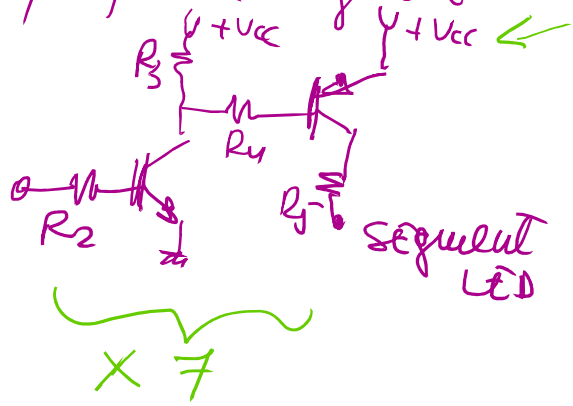
ANOD COMUN

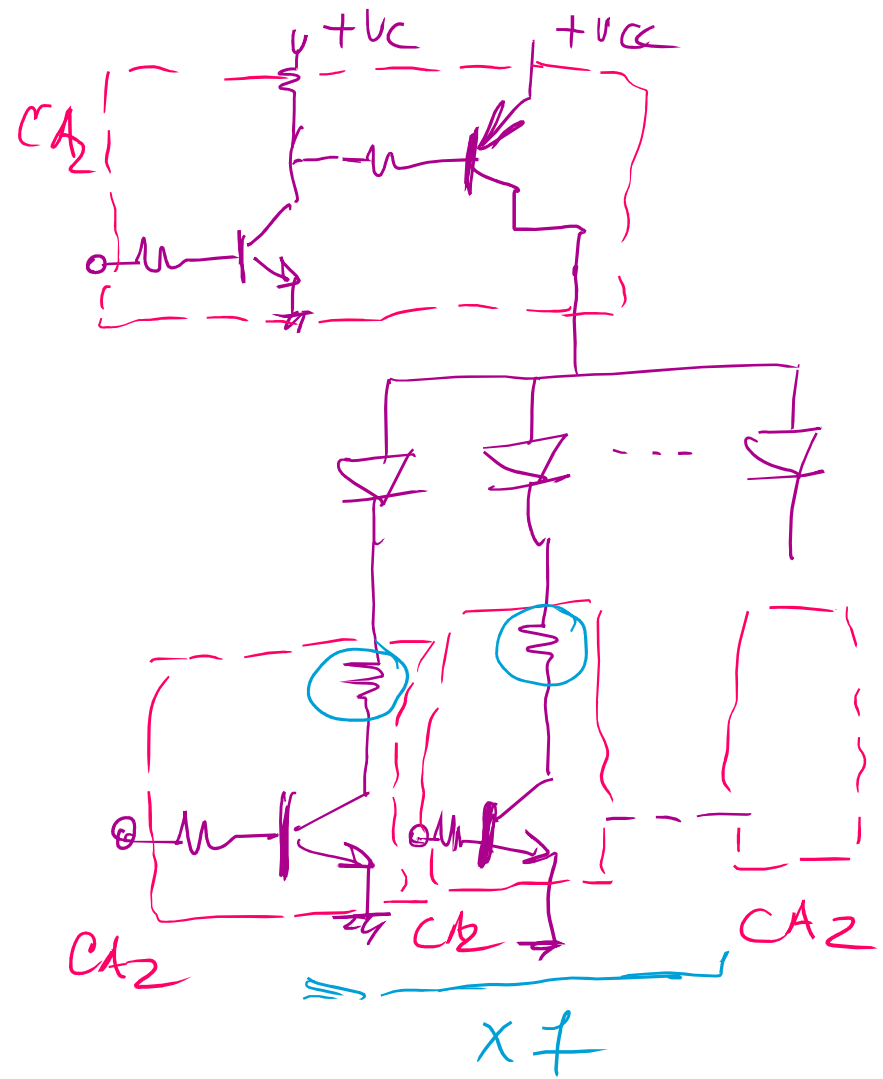
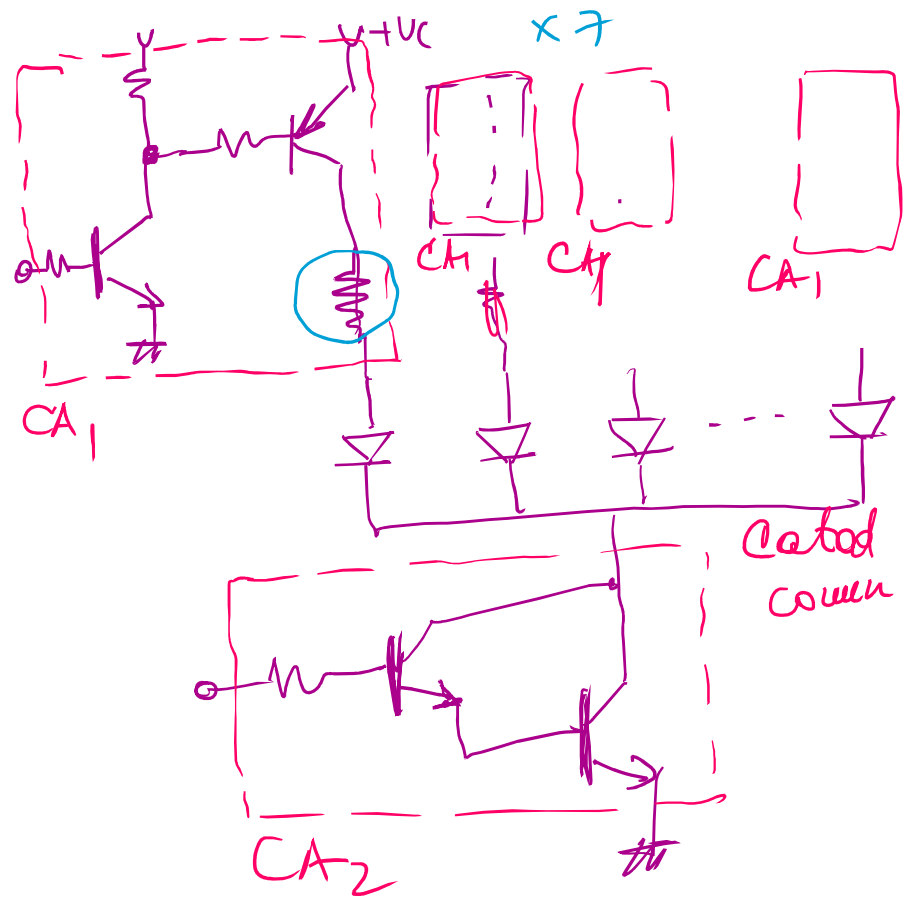


Circuit amplifcare curent pe segmente (CA₁)

CATOD COMUN

pe fiecare segment



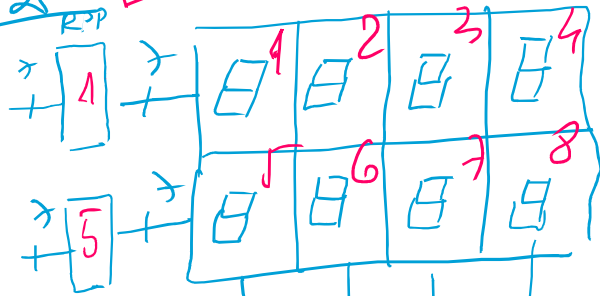


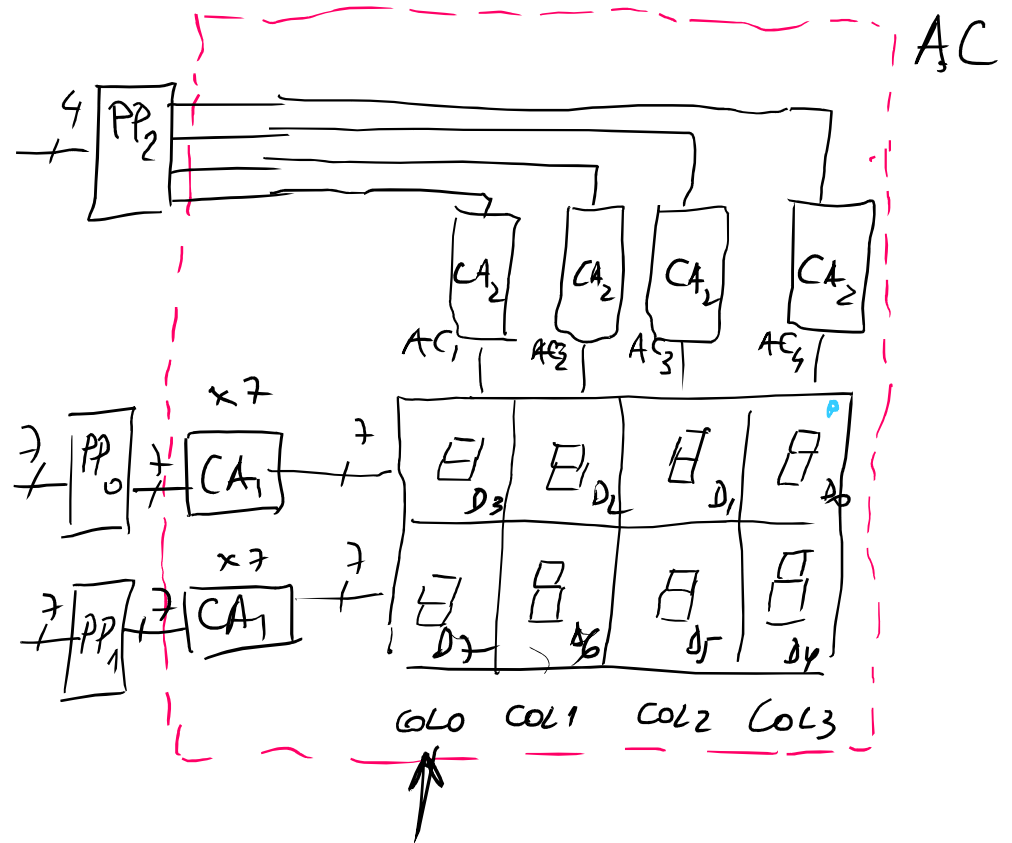
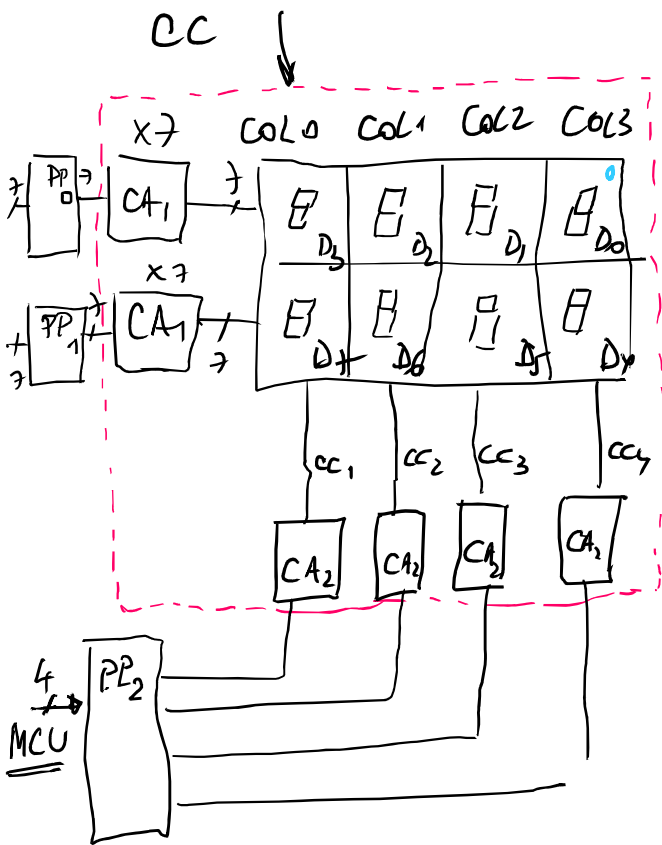
Dispunere celule afroaj

caz 1



caz 2 $\square$ CA1 $\rightarrow 2 \times 4$





END 4 mar 2026

D ₃	D ₂	D ₁	D ₀
D ₇	D ₆	D ₅	D ₄

(comando
electrice)

D ₃	D ₄	D ₅	D ₆	D ₇	D ₈	D ₉	D ₀
----------------	----------------	----------------	----------------	----------------	----------------	----------------	----------------

t_0 1 3 5 9 7 8 4 0
 t_1 3 5 9 7 8 4 0 1
 t_2 5 9 7 8 4 0 1 3

RAM

CONV.
 BIN
 7 SG

⇒
 programme
 a fisare

D ₀
D ₁
D ₂
D ₃
D ₄
D ₅
D ₆
D ₇
DBIN ₀
DBIN ₁
DBIN ₂
DBIN ₃
DBIN ₄
DBIN ₅
DBIN ₆
DBIN ₇

⇒
 a fisare
 conect

1
 00000001
 00000010
 00000011
 00000100

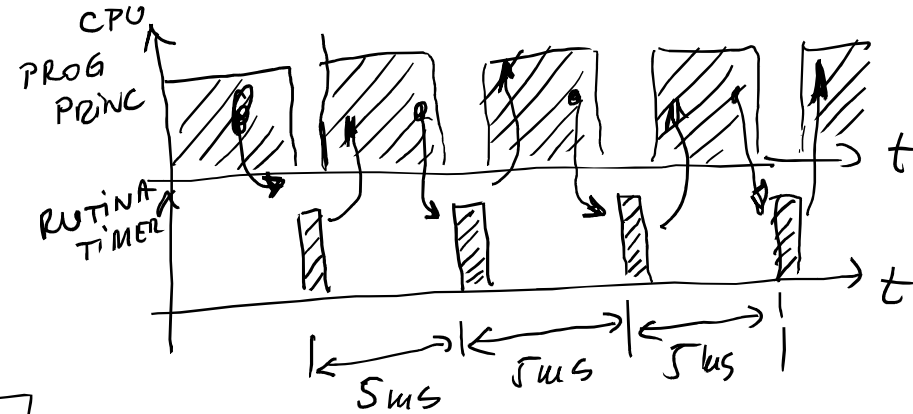
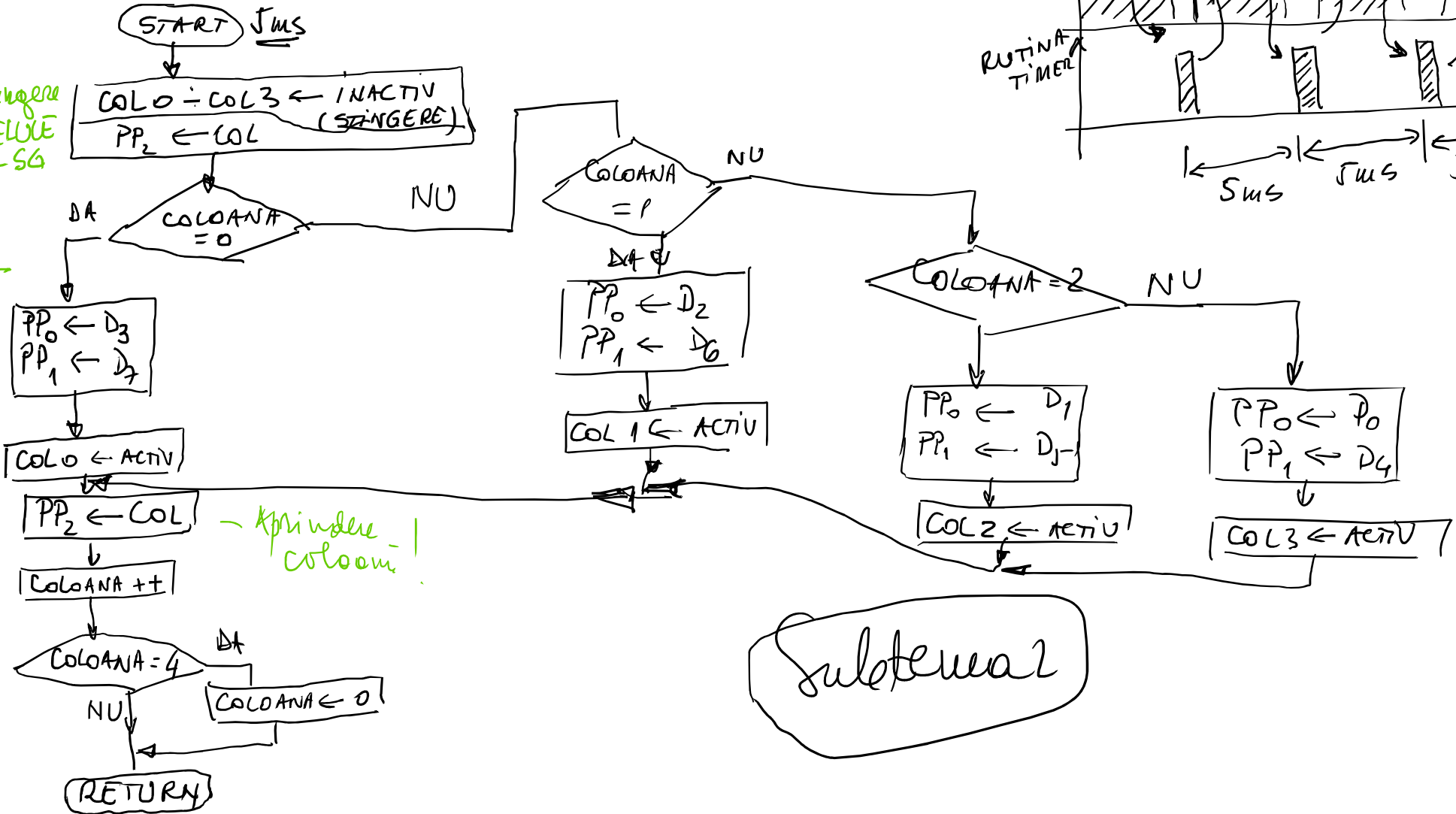
INC
 DIGIT + 1

COL ⇒ COL 0, COL 1, COL 2, COL 3

RUTINA TRATARE
INTERRUPTURE TIMER
(SEMNAUZARE TRECERE 5ms)

stingere
celule
7 SA

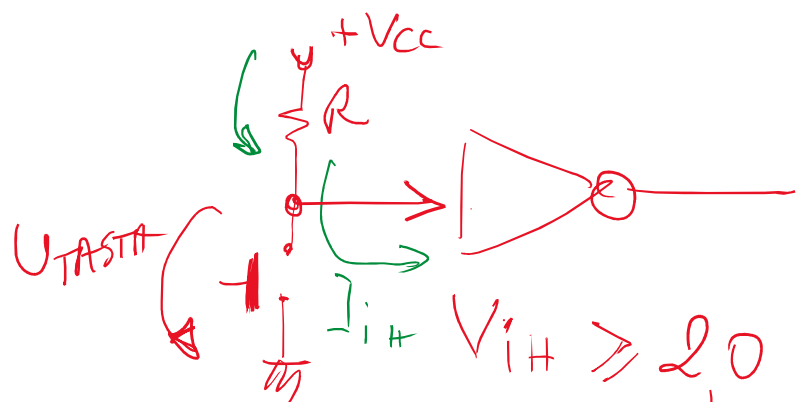
aprinde
coloani!



subternea?

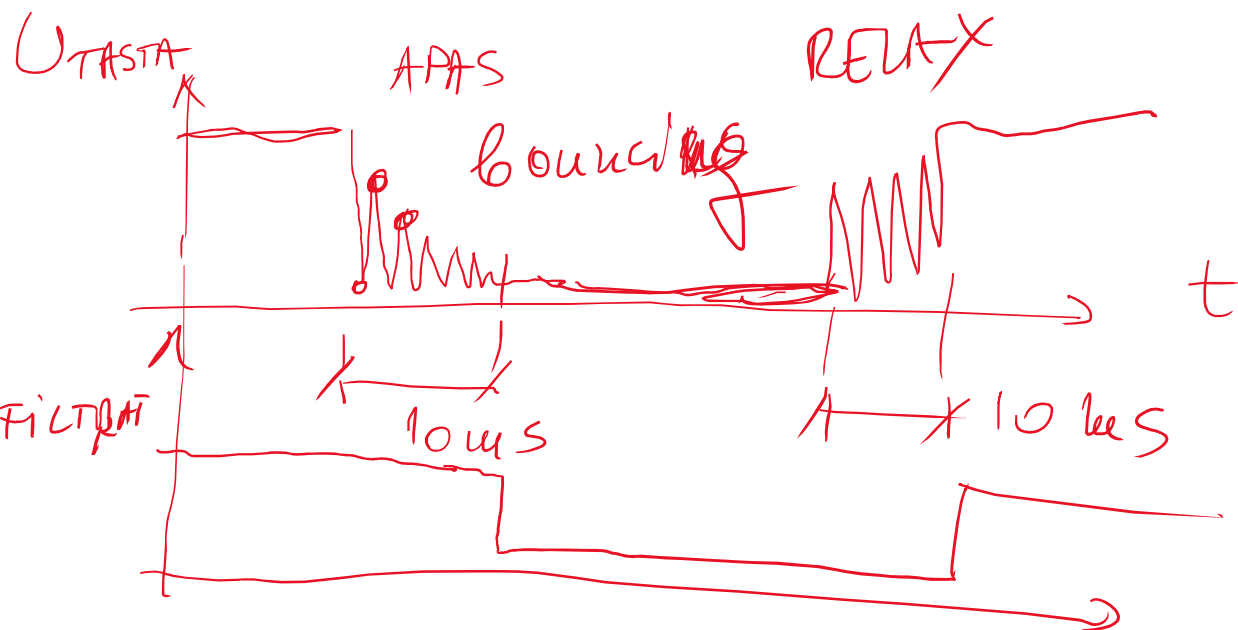
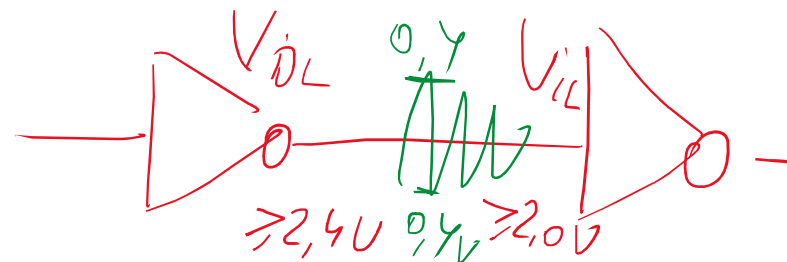
Nume variabila	Semnificatie	Valoarea de initializare	Mod de reprezentare	Adresa de stocare in memorie RAM
D0-D7	Imagini SW format 7 sg pentru celulele LED	7f(h) – CC; 00(h) – AC (segmente aprinse)	octet (8) – 7 biti utili / octet	30(h) – 37(h)
DBIN0-DBIN7	Imagini SW format binar (BCD) pentru celulele LED	08(h)	octet (8) – 4 biti utili / octet	38(h) – 3f(h)
COL0-COL3 (COL)	Imagini SW pentru comenzi pe coloanele comune	01(h) – AC; 0e(h) - CC	4 biti – accesibili individual	20(h) – zona adresabila pe Bit (00h, 01h, 02, 03h)
COLOANA	Indica ce coloana se actioneaza	03 – 00 (4 valori)	Octet	40(h)

END 11 mar 2026

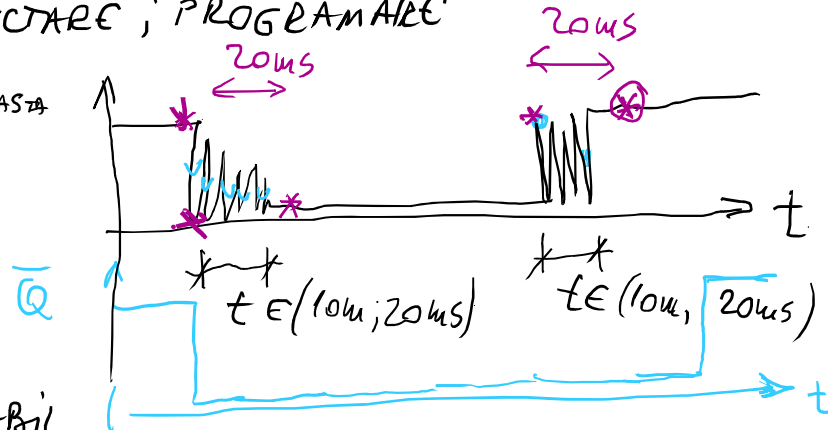
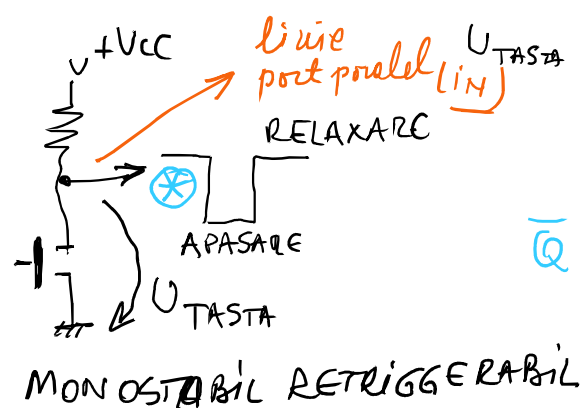


$$V_{IH} \geq 2,0V - V_{OH} \geq 2,4V$$

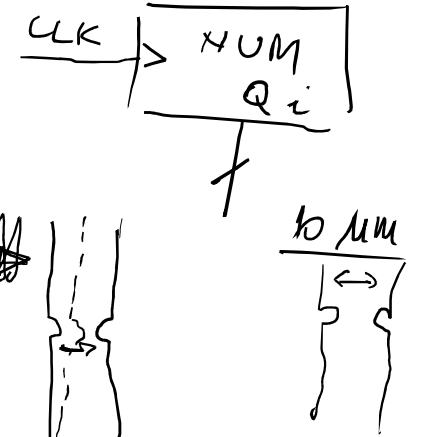
$$V_{IL} \leq 0,8V - V_{OL} \leq 0,4V$$



TASTATURA, CONECTARE, PROGRAMARE

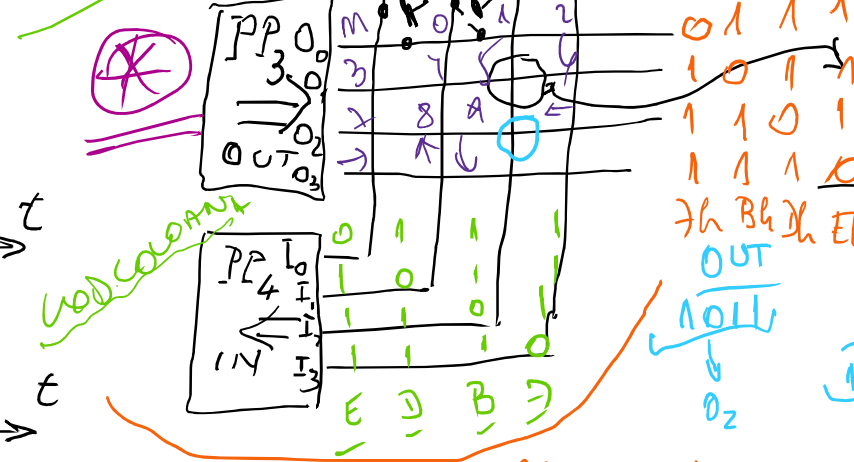


bounce debouncing



COMANDA MULTIPLEXATA TASTATURA

COD. BINEIERE



SUBTEMA 3 (PARTE HW TASTATURA)

20K - 1M

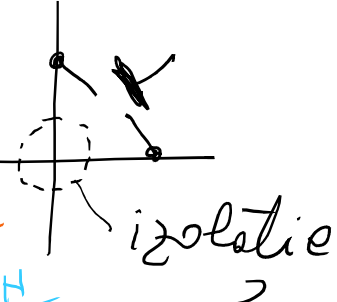
5V - 250 μV

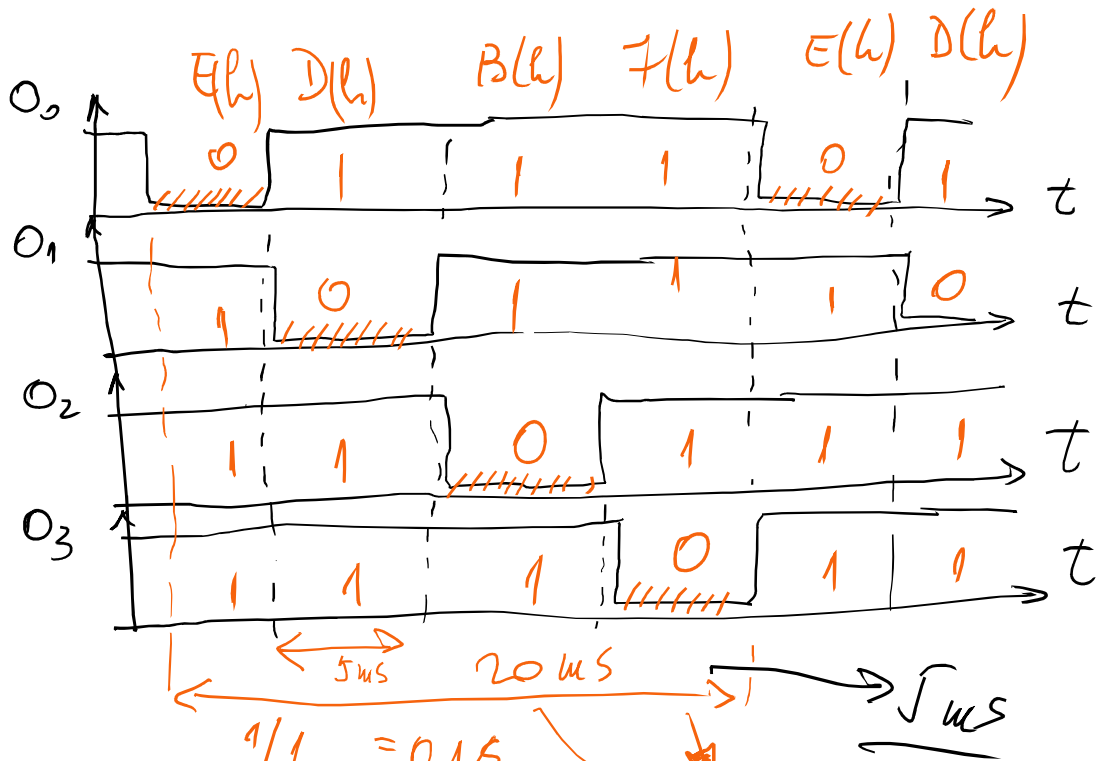
folosire optimale linii PP

14 TASTE

2 x 7 = 9#

4 x 4 = 8#

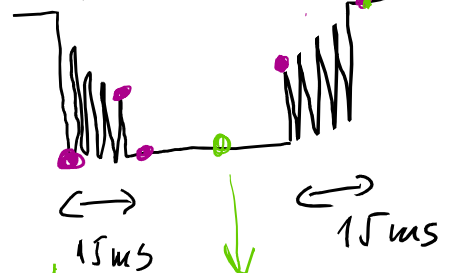




$$1/10 = 0,15$$

(100ms)

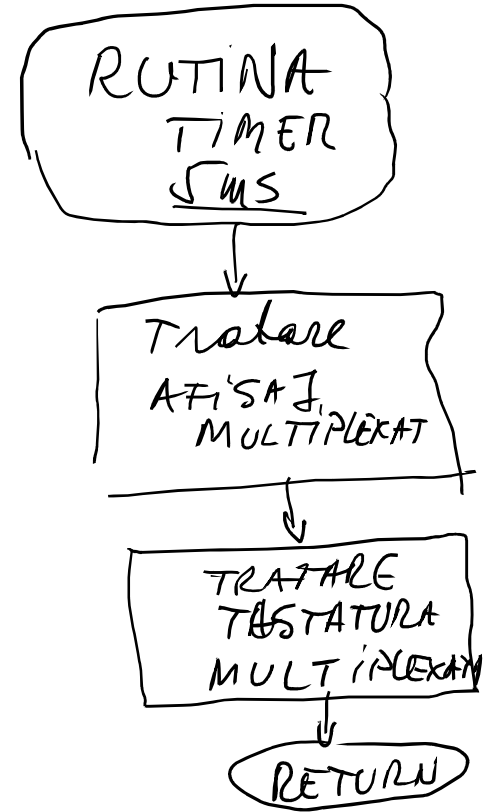
APASARE SI RELAXARE TASTA



APAS | TASTA_APAS | RELAX |

Proces de baleiere a tastaturii -

$E \rightarrow D \rightarrow B \rightarrow 7 \rightarrow E$



TIMER 5ms

TRATARE
AFISAJ
MULTIPLEXAT

TRATARE
TASTATURA
MULTIPLEXATA

RETI

SEMAFOR_TASTA

- indica
tasto
apreso
corect

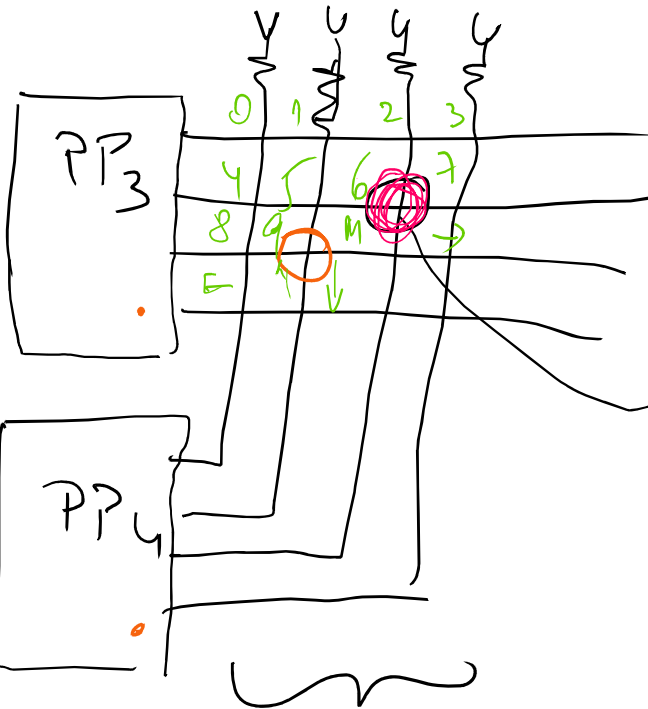
COD_BALEIA]

COD_TASTA_COL

{
tasto
apreso

=> progr.
principal

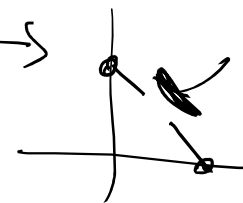
cod
baieia
 $\in \{07h, 08h, 09h, 0Eh\}$



COD_TASTA_COL
(4b)

locus

COD_BALEIA]
(4b)



COD_BALEIA]
(4b)

COD_TASTA_COL
(4b)

1 0 1 1
B

1 1 0 1
D

0
B)h

07h 08h
0 1 1
1 0 1
1 1 0
1 1 1

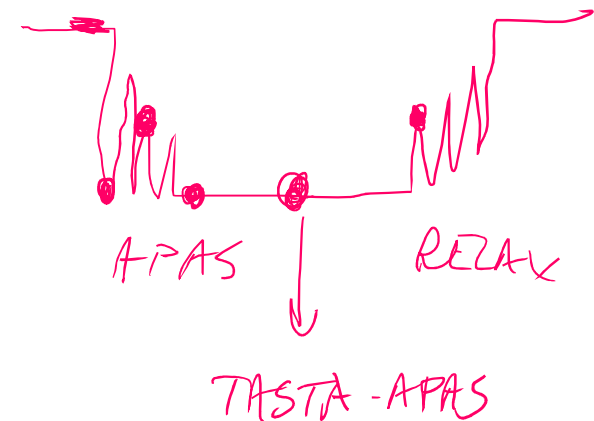
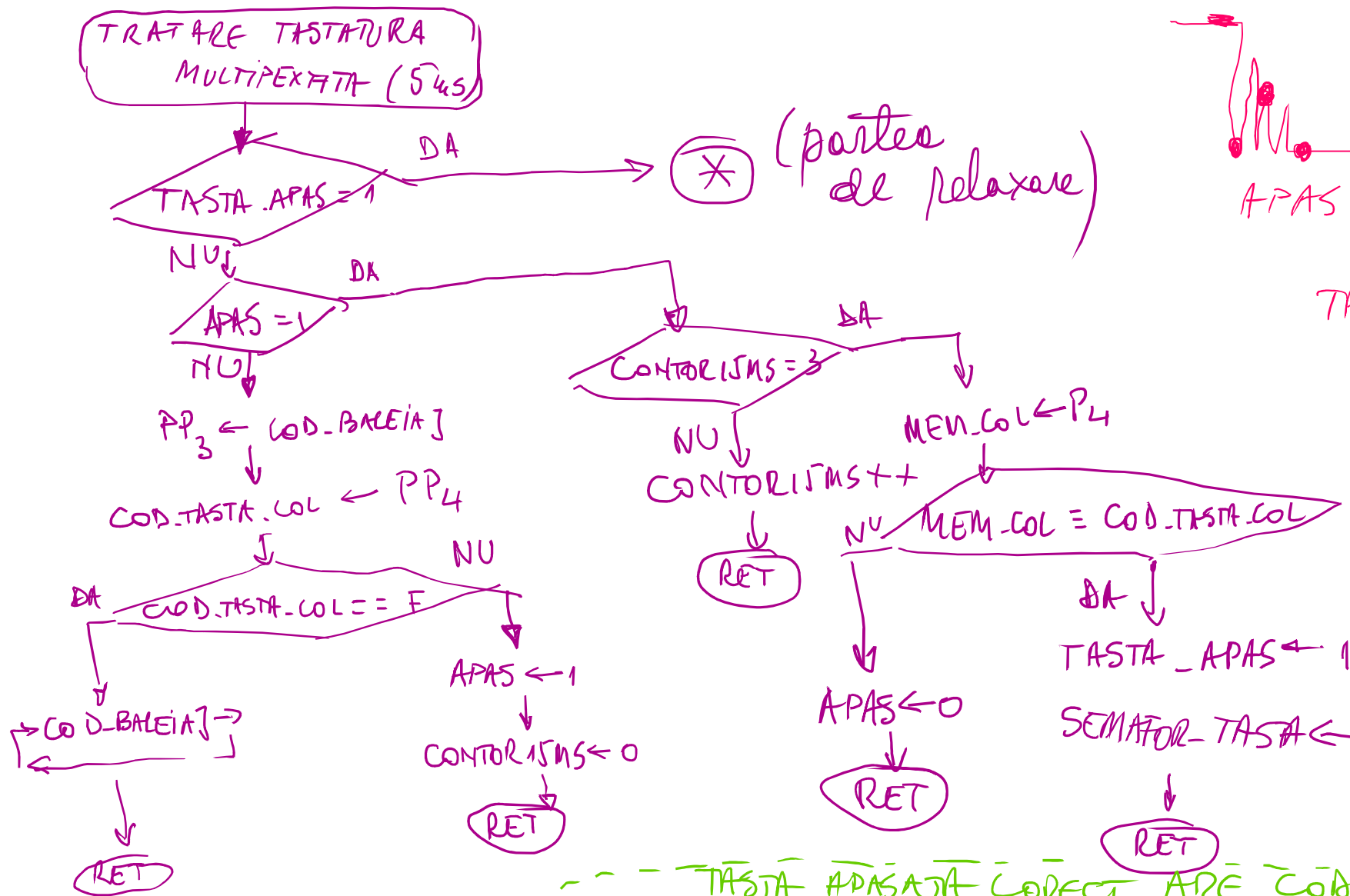
⊗ La lucrul cu o tastatură este necesar:

- A - identificarea momentului episozi unei teste
- B - realizarea unei temporizări (15 ms) pentru evitarea bouncing-ului
- C - reverificarea stării de apăsare a testei respective (0)
- identificarea momentului relaxării unei teste
- realizarea unei temporizări (15 ms) pentru evitarea bouncing-ului
- reverificarea stării de relaxare a testei (1)

⑦ Prelucrarea unei taste apăsate de către o altă componentă soft necesită:

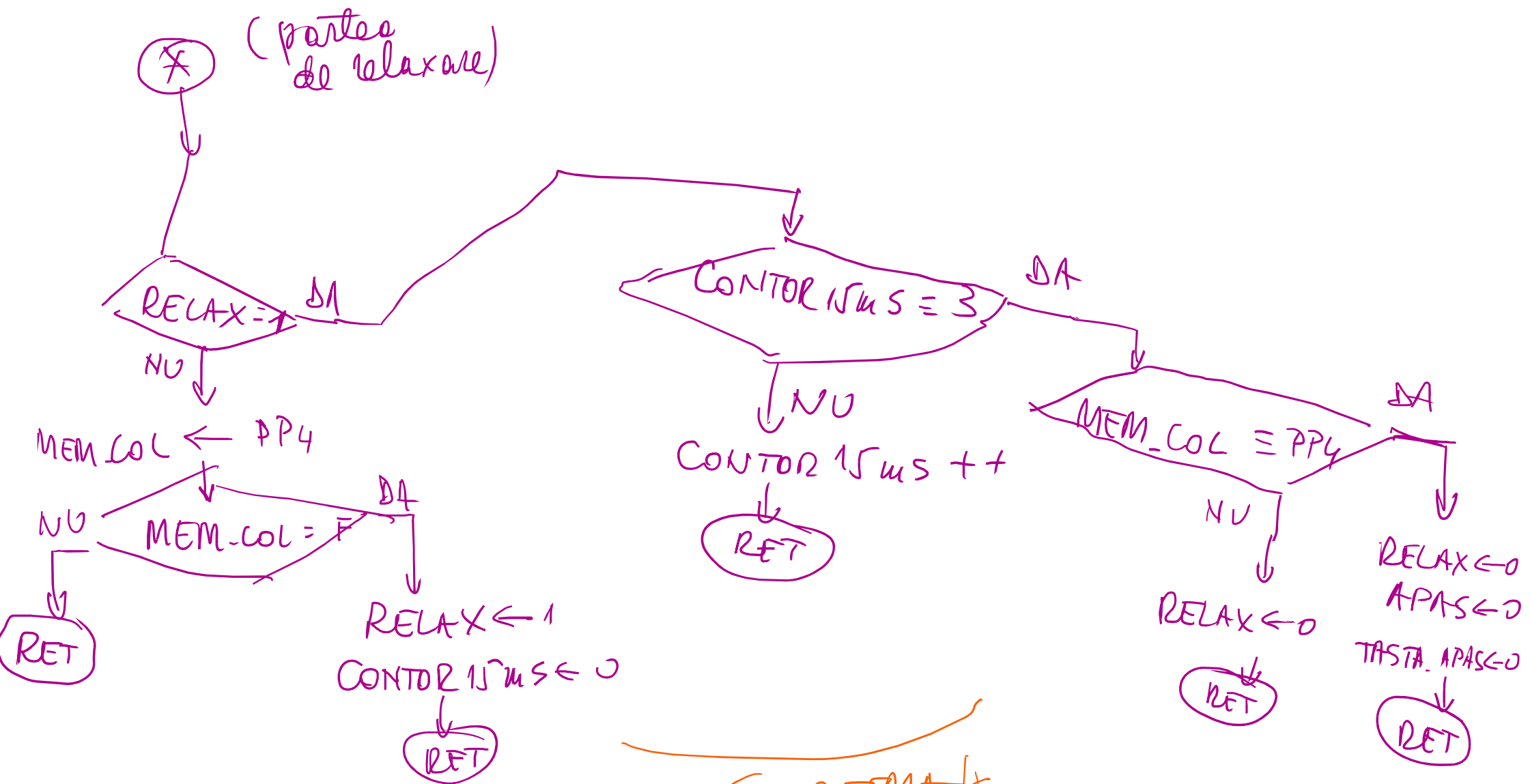
- semnalizarea unei taste corect apasate (presii A, B, C)
(variabila SEMAFOR-TASTA)
- codul de identificare a tastei apasate (variabilele obtinute prin concatenarea cod de balieare-pe linii si cod citit coloana)
COD - BALEIAJ COD - TASTA - COL

END 18 mar 2026



--- TASTA APASATA CORECT ARE CODUL DE IDENT.
FORMAT AIN COD.BALEIA] SI COD.TASTA.COL !

INBICA
CATRE
PROG.PRINC
APASARE
CORECTA TASTA!



SUBTEMA 4

(SW - TASTATURA MUX.)

Manual de utilizare al tabelor

Subtema 5

- ① Tabelul are 2 moduri de funcționare
 - 1) mod 0 - operare normale

4 mod 1 - programare

Modul se scrie prin apăsarea tastei M

- ② În modul operare normale

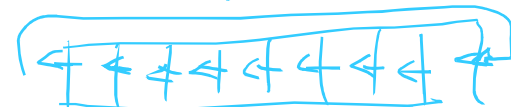
- textul afișat se deplasează
- cifre stâng și dreapta viteze de deplasare indicate în funcție
- supraveghere taste M
- și curenții mod
- nici o altă tastă nu acționează în modul 0 !

- ③ Modul programare
 - = textul afișat rămâne fix
 - = primul digit (dreapta) va clipi
 - Cu timpul indicat în funcție
 - = supraveghere taste apăsate
 - 0, ..., 9, $\leftarrow \uparrow \rightarrow$, M

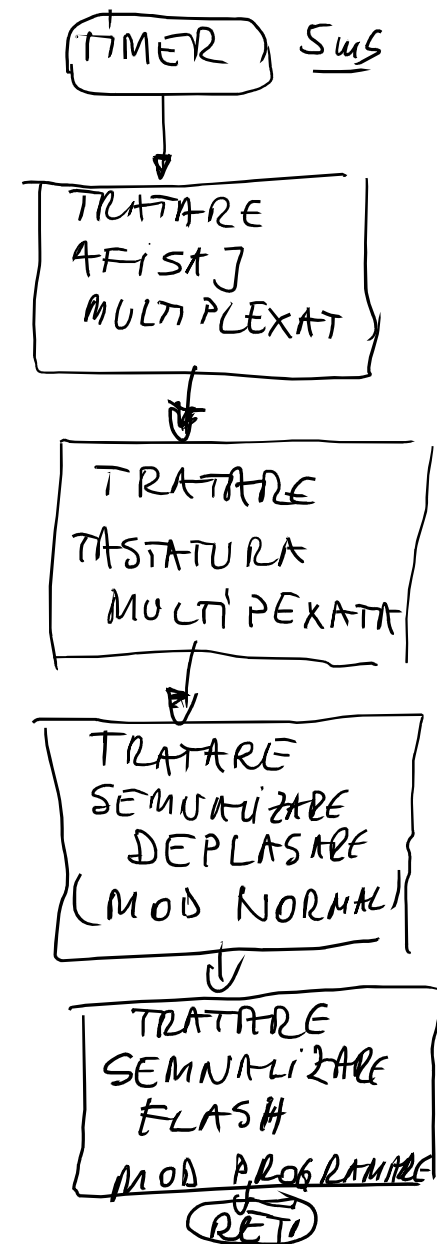
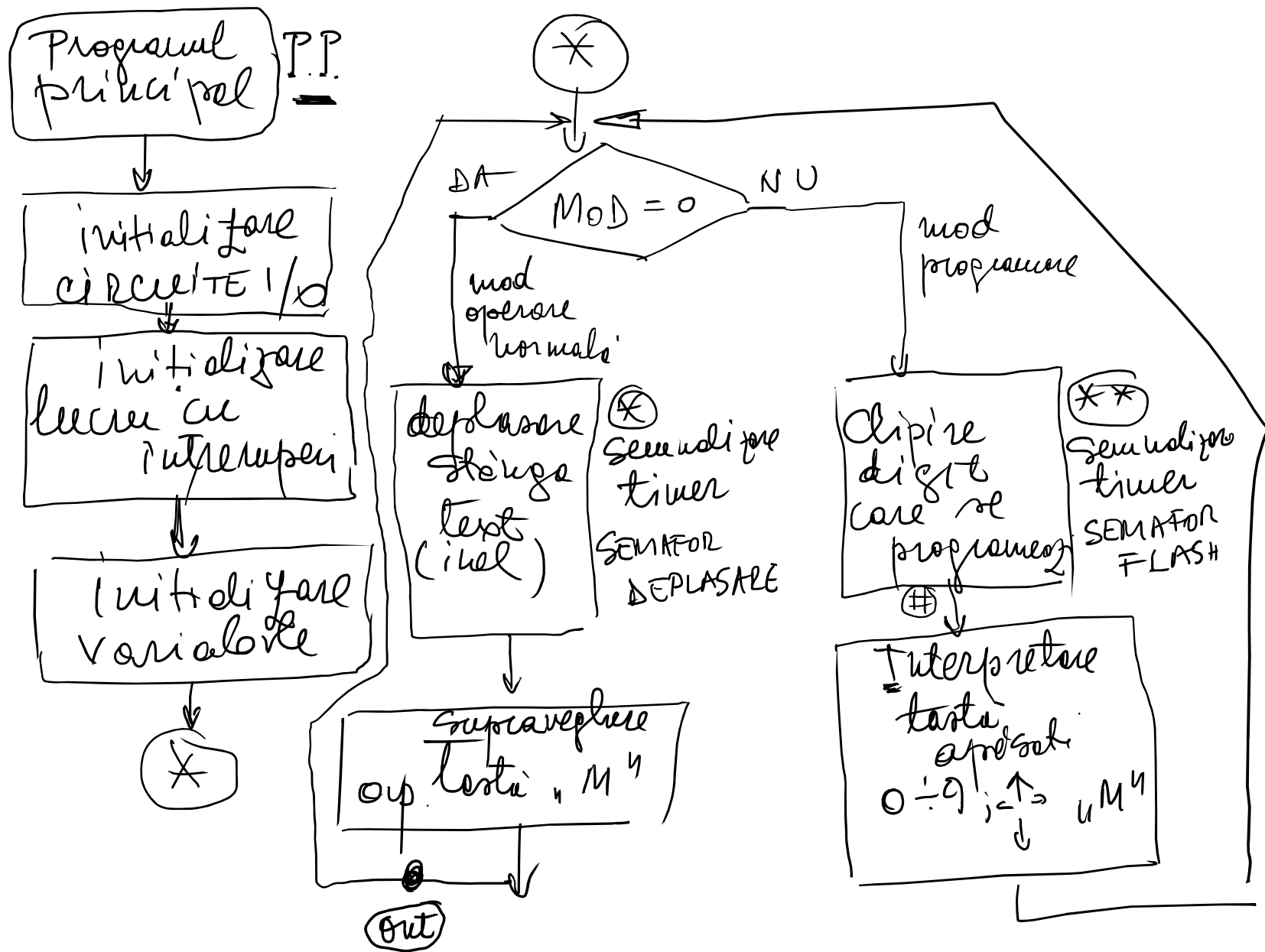
= la tastele 0 ÷ 9 → se programează digitul cu valoarea

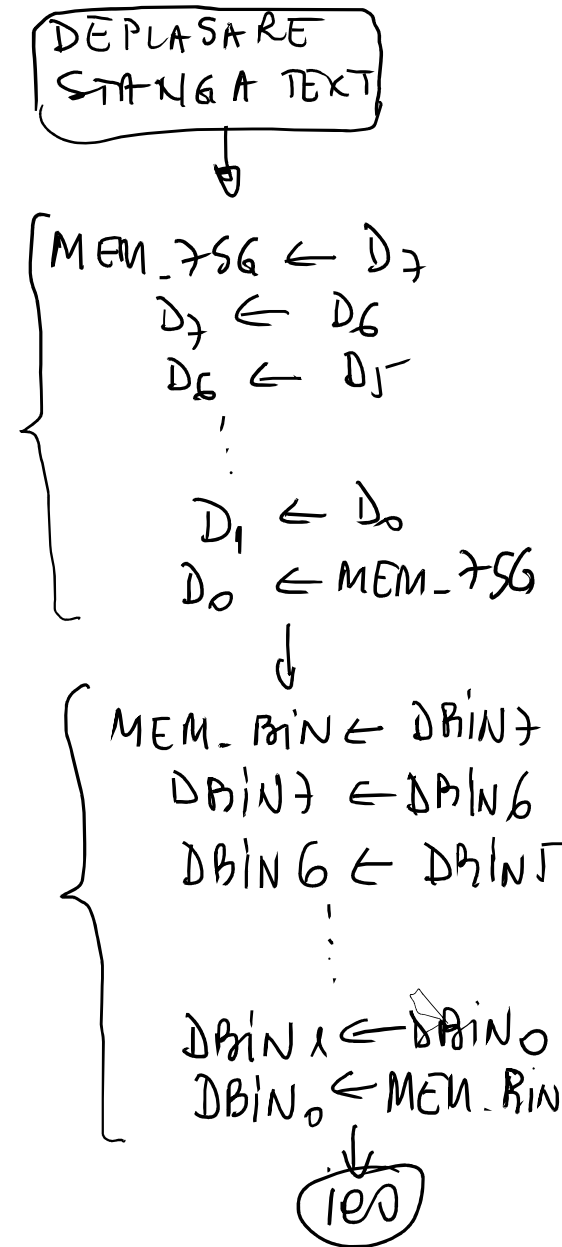
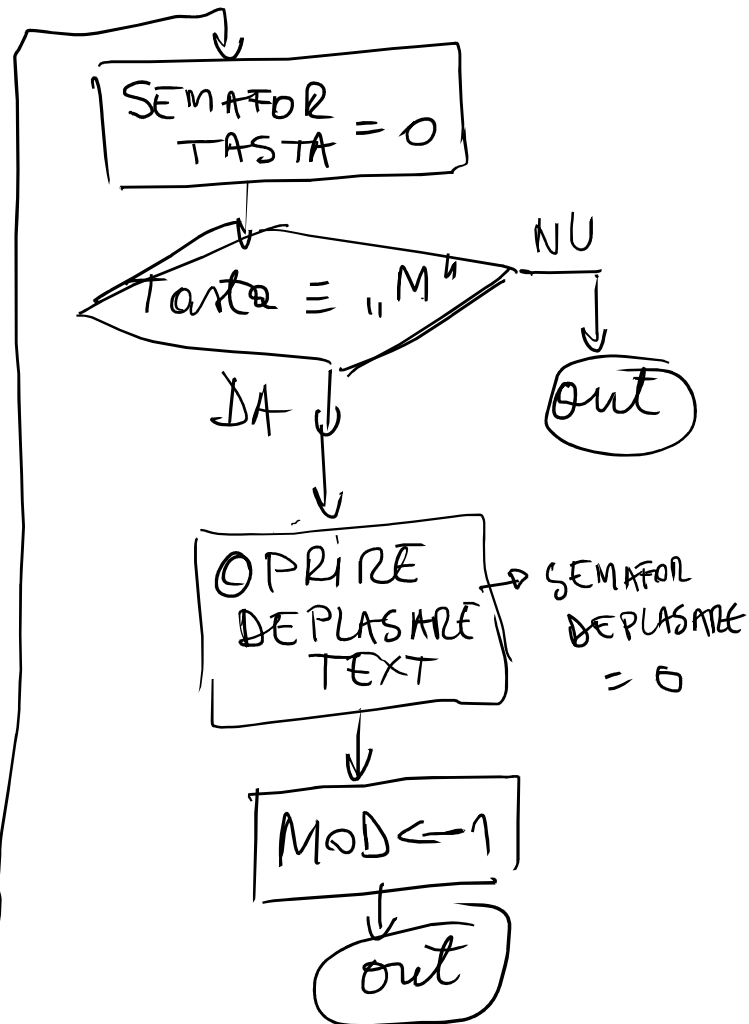
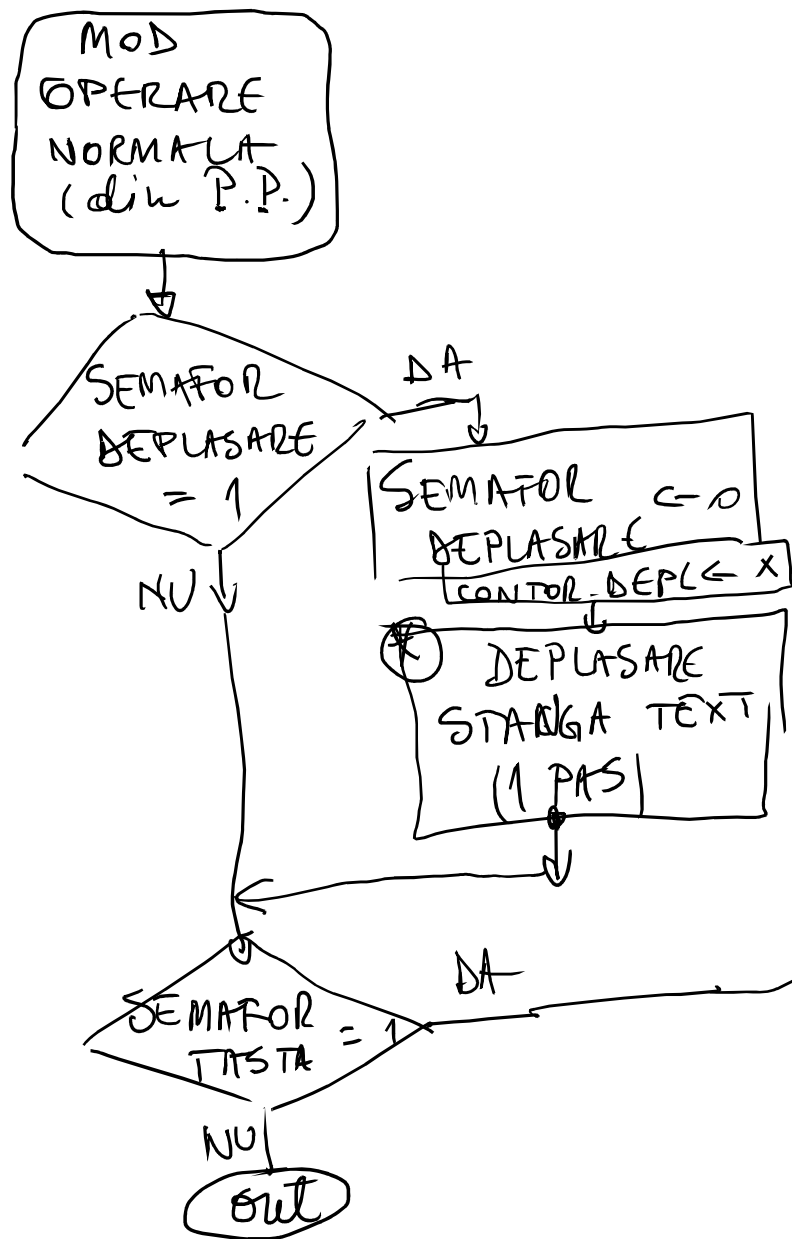
= cu tastele $\uparrow$ - incrementare
 $\downarrow$ - decrementare

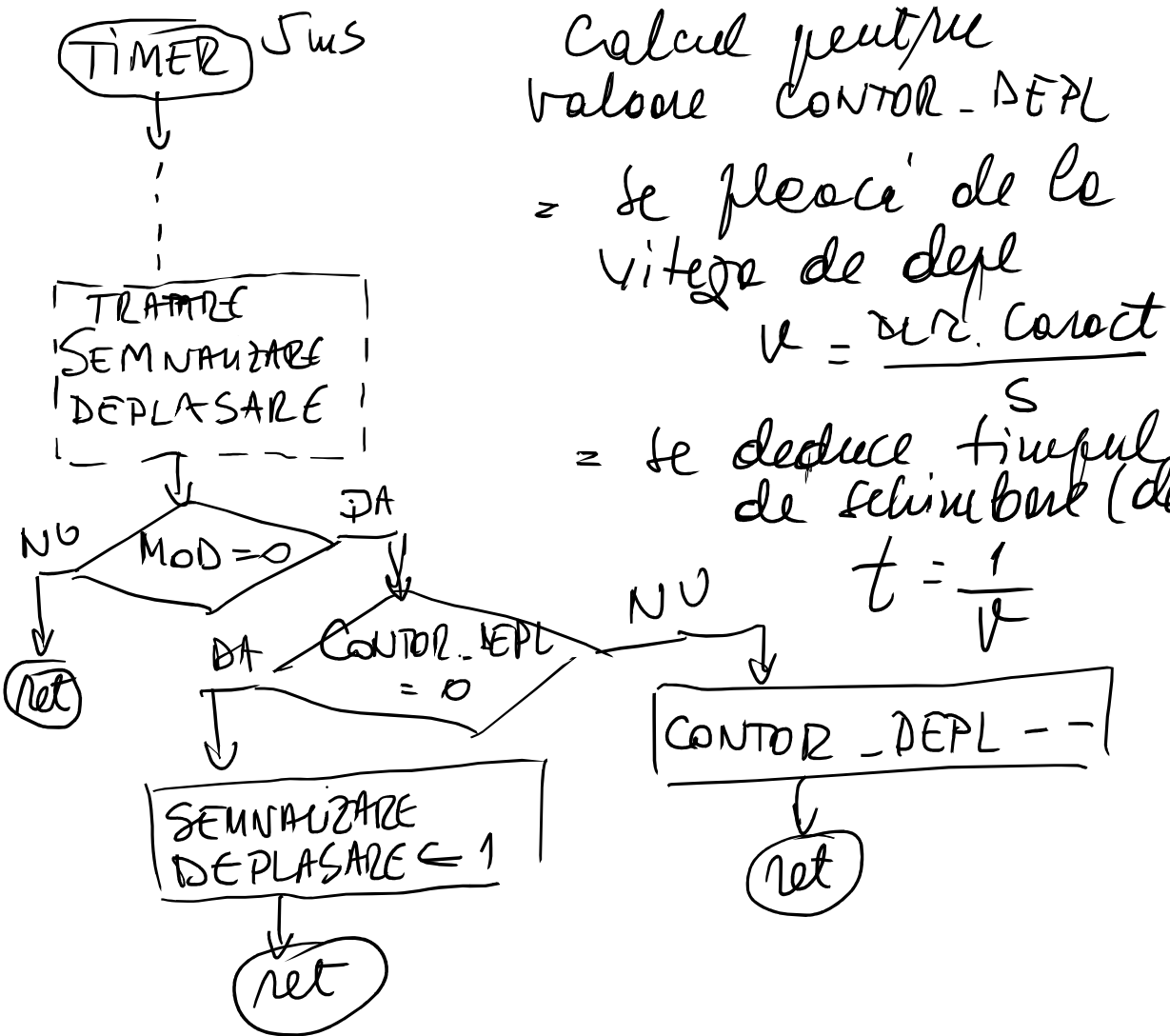
= cu tastele $\leftarrow \Rightarrow$ aleg următorul digit care se programează $\leftarrow$



END 25 mar 2026







Calcul pentru
valoare CONTOR-DEPL
= se pleacă de la
viteza de depl
 $v = \frac{nr. \text{caract}}{s}$

= se deduce timpul
de schimbare (deplasare)
 $t = \frac{1}{v}$

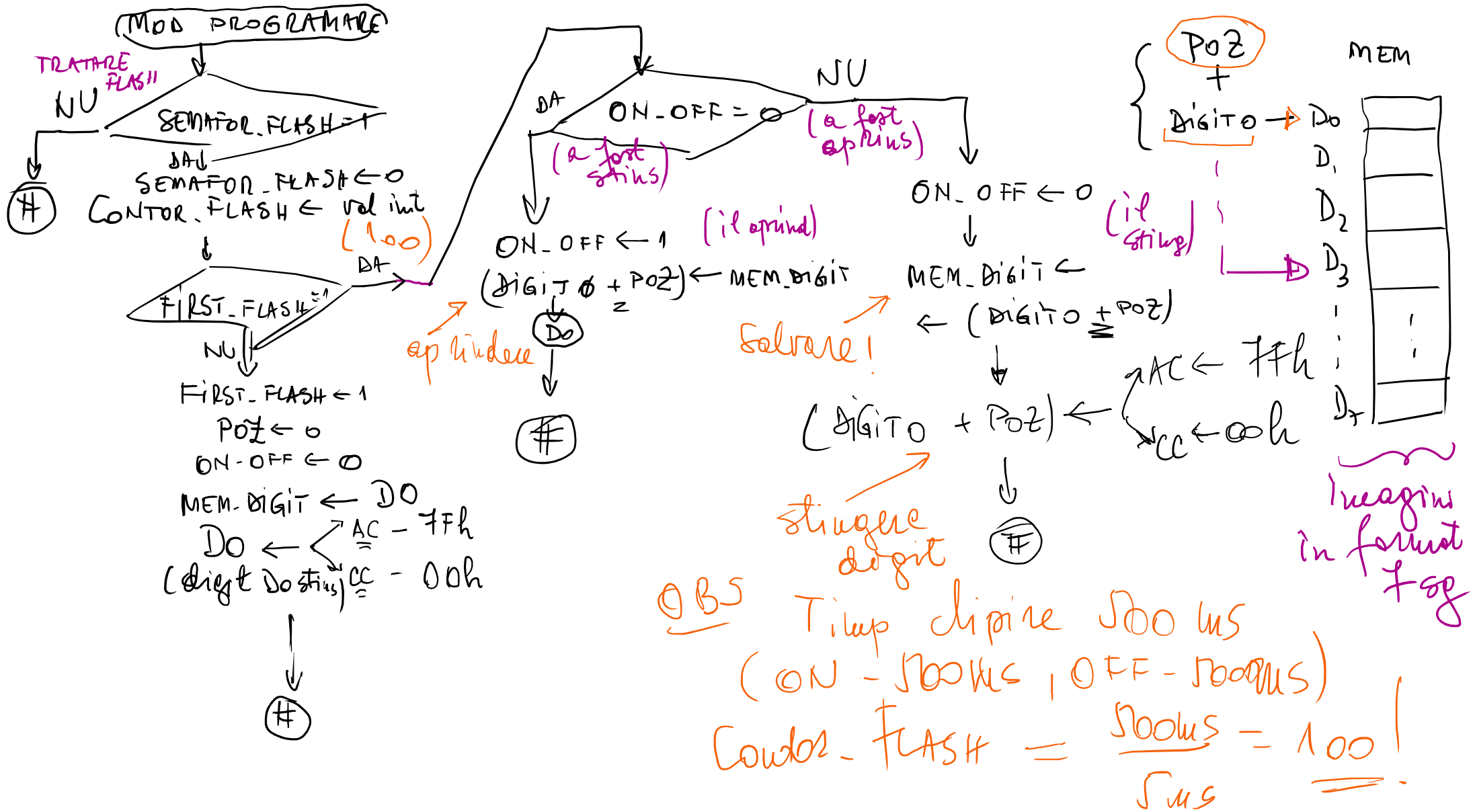
= valoarea
CONTOR-DEPL
 $= \frac{1}{v}$
 $\frac{5ms}{0.005s}$
etalonul
caz care maror
timpul

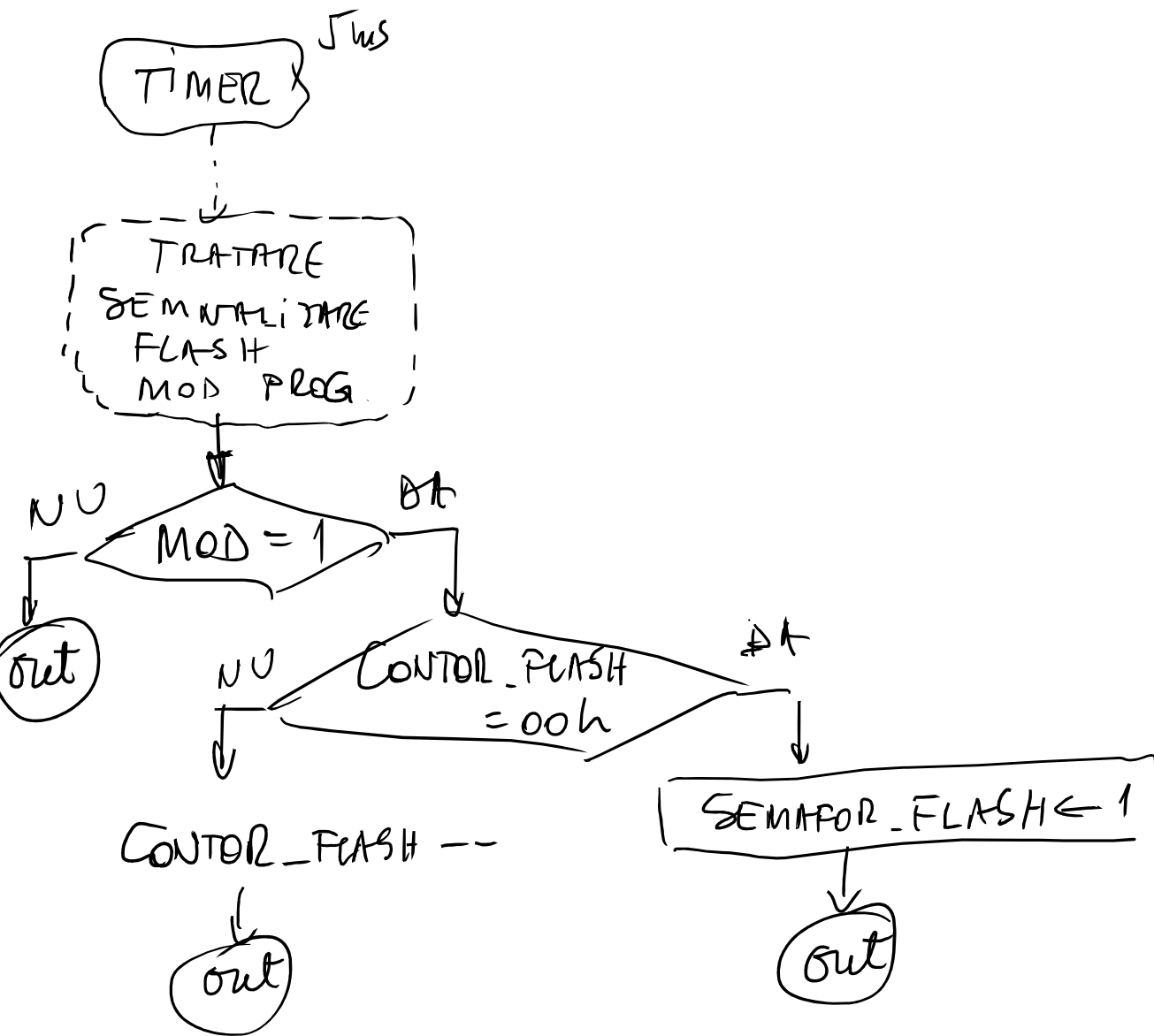
PROBLEMA

- verificarea operații
cu valoarea CONTOR-DEPL

ex 4 caract/s
 $v = 4 \frac{1}{s}$
 $t = \frac{1}{4}$
 $CONTOR = \frac{1.000}{4 \cdot 5} = 50 \text{ (V)}$

END 1 apr 2026





END 8 apr 2026